Angewandte Chemie
Eine Zeitschrift der Gesellschaft Deutscher Chemiker

Supporting Information

Diaminodicyanoquinones: Fluorescent Dyes with High Dipole Moments and Electron-Acceptor Properties

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1. General Information

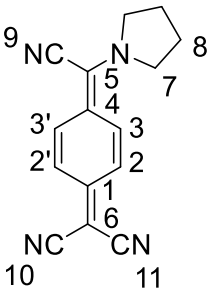
All reagents were purchased from commercial sources and used without further purification. Dry solvents were purchased from Acros Organics. ALUGRAM Xtra SIL G/UV_{254} plates by Macherey-Nagel were used for thin-layer chromatography. Isolation of products by chromatography was performed with silica from Macherey-Nagel Silica 60 M (0.04-0.063 mm).

NMR spectra were recorded on a JOEL ECX 400 (^{1}H 400 MHz, ^{13}C 101 MHz), JEOL Eclipse+ 500 (^{1}H 500 MHz, ^{13}C 126 MHz) and BRUKER AVANCE 700 (^{1}H 700 MHz, ^{13}C 176 MHz) spectrometer at 25 °C. The chemical shifts δ are calibrated on the respective solvent peak as internal standard. All shifts are reported in ppm and NMR multiplicities are abbreviated as s (singlet), d (duplet), t (triplet), m (multiplet). Coupling constants J are reported in Hz. UV/Vis spectra were recorded on a Cary 50 Bio photo spectrometer (Varian). Fluorescence spectra were recorded on a LS 50 B luminescence spectrometer from PerkinElmer. UV/Vis and Fluorescence spectra were measured in quartz glass cuvettes with 1 cm path length. IR Spectra were recorded on a FT/IR 4100 spectrometer from JASCO. Elemental analysis was performed on an VARIO EL from Elementar.

2. Synthetic Procedures

The synthetic procedure was employed from Ref. [1] and used with only minor modifications.

2-(4-(cyano(pyrrolidin-1-yl)methylene)cyclohexa-2,5-dien-1-ylidene)malononitrile (8)



To a stirred warm solution of TCNQ (250.0 mg, 1.22 mmol, 1 eq.) in acetonitrile (20 mL), pyrrolidine (69.7 mg, 0.98 mmol, 0.8 eq.) was added in one shot. The solution turned green and then purple. After stirring for 4 hours at 70 °C the solution was cooled to room temperature and then stored in the fridge for 3 days. The precipitate was filtered and washed with cooled acetonitrile (3x 5 mL) to yield the product as fine purple crystal needles (232.4 mg, 0.93 mmol, 76.5%).

^{1}H NMR (400 MHz, DMSO-d_6, RT): δ (ppm) = 7.73 (dt, ^{4}J = 0.7Hz, ^{3}J = 7Hz, 2H, 3), 7.81 (dt, ^{4}J = 0.7Hz, ^{3}J = 7Hz, 2H, 2), 4.11 (bs, 4H, 7), 2.08 (m, 4H, 8)

^{13}C NMR (100 MHz, DMSO-d_6, RT): δ (ppm) = 153.95 (1C, 6), 137.76 (1C, 4), 134.32 (2C, 2), 120.11 (2C, 3), 118.92 (1C, 10), 116.78 (1C, 11), 113.09 (1C, 9), 57.40(1C, 6), 50.92 (2C, 7), 25.60 (2C, 8)

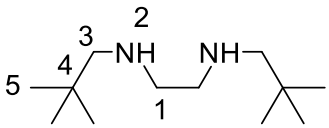
FT-IR (ATR) $\tilde{\nu}$ (cm^{-1}): 2190 (m), 2164 (s), 1611 (s), 1536 (m), 1474 (w), 1381 (s), 1341 (m), 1207, (s), 861 (s), 823 (s), 727 (m), 650 (s)

UV/Vis (ACN) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 275 (6300), 480 (13000)

MS (EI): m/z = 249.1 (20); 248.0 (100)[MH]$^{\cdot+}$; 221.1 (15); 179.1 (17); 154.1 (20)

EA: $C_{15}H_{12}N_4$; calc.: C, 72.56; N, 22.57, H, 4.87; meas.: C, 72.74; N, 21.57; H, 4.89

N1,N2-dineopentylethane-1,2-diamine (9)



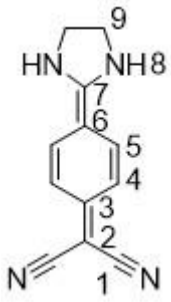
Ethylenediamine (180.0 mg, 3.0 mmol, 1.0 eq.) and Pivalaldehyde (593.3 mg, 6.9 mmol, 2.3 eq.) were stirred in dry MeOH (20 mL) under argon atmosphere for 12 hours. $NaBH_4$ (453.2 mg, 12.0 mmol, 4 eq.) was added stepwise and the reaction mixture was refluxed for 2 hours. The reaction was quenched with water (10 mL), and the aqueous phase was extracted with hexane (3x 15 mL). The combined organic phases were dried over $MgSO_4$ filtrated and evaporated under reduced pressure. The crude product was purified by column chromatography (Ethyl acetate / MeOH 90 / 10 with 1%v/v 7M NH_3 in MeOH) to yield a white solid (520 mg, 2.6 mmol, 86.7%, RF = 0.2).

^{1}H NMR (400 MHz, DMSO-d_6, RT): δ (ppm) = 2.68 (s, 4H, 1), 2.30 (s, 4H, 3),0.90 (s, 18H, 5)

^{13}C NMR (100 MHz, DMSO-d_6, RT): δ (ppm) =62.32 (2C, 1), 49.91 (2C, 3), 31.58 (2C, 4), 27.77 (6C, 5)

FT-IR (ATR) $\tilde{\nu}$ (cm^{-1}): 2962 (s), 2866 (w), 2811 (w), 1541 (m), 1478 (s), 1456 (s), 1364 (m), 910 (s), 760 (m), 733 (s)

2-(4-(imidazolidin-2-ylidene)cyclohexa-2,5-dien-1-ylidene)malononitrile (1)



Ethylenediamine (26.6 mg, 0.44 mmol, 1 eq.) was added to a 40 °C warm solution of **8** (110 mg, 0.44 mmol, 1 eq.) in acetonitrile (10 mL). The solution turned deep green immediately and to yellow in the next minutes. The solution was stirred at 70 °C for 4 hours and then cooled to room temperature. The precipitate was filtered off and washed with cooled acetonitrile (3x 5 mL) to yield the product as a yellow fine-grain powder (73.4 mg, 0.35 mmol, 78.8%).

^{1}H NMR (500 MHz, DMSO-d_6, RT): δ (ppm) = 9.76 (s, 2H, 8), 7.57 (d, ^{3}J = 5.0 Hz, 2H, 4), 6.82 (d, ^{3}J = 5.0 Hz, 2H, 5), 3.87 (bs, 4H, 9)

^{13}C NMR (125 MHz, DMSO-d_6, RT): δ (ppm) = 164.73 (1C, 7), 150.53 (2C, 1), 129.48 (1C, 3), 124.40 (2C, 4), 117.90 (2C, 5), 108.73 (2C, 6), 44.36 (2C, 9), 35.42 (2C, 2)

FT-IR (ATR) $\tilde{\nu}$ (cm^{-1}): 3084 (w), 3022 (vw), 3009 (vw), 2183 (s), 2144 (vs), 1594 (s), 1580 (s), 1507 (s), 1332 (s), 1282 (s), 1236 (w), 1199 (m), 952 (w), 873 (s), 747 (m)

UV/Vis (DMSO) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 405 (22800)

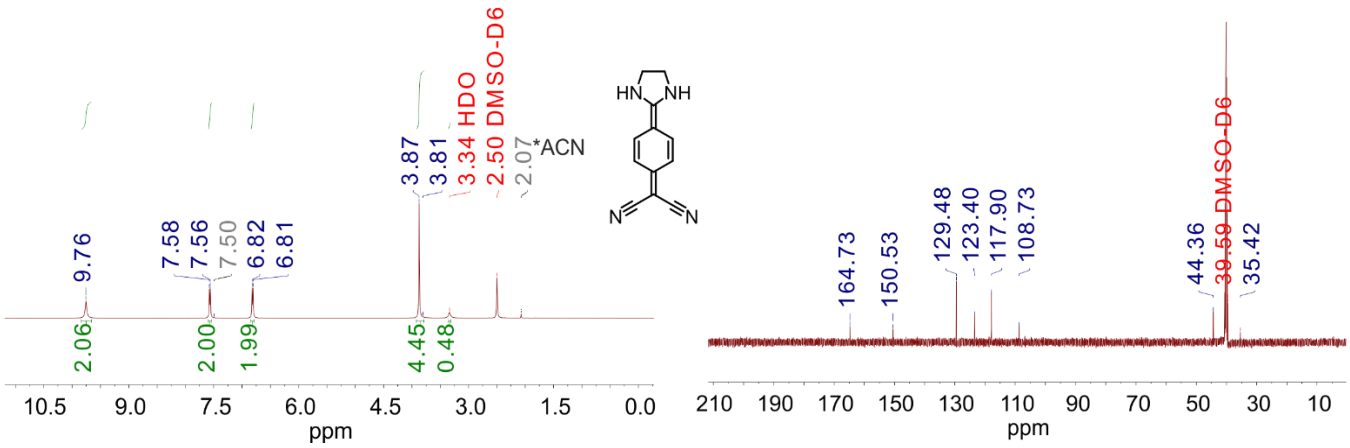
UV/Vis (ACN) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 357 (4000), 409 (4000)

UV/Vis (DMF) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 363 (13000), 410 (18000)

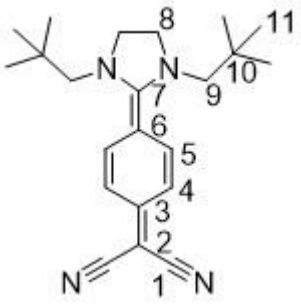
UV/Vis (Methanol) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 395 (36875)

MS (EI): m/z = 210.1 (100)[MH]$^{\cdot+}$; 209.1 (50); 182.1 (40); 181.1 (100); 155.1 (15); 141.0 (15); 114 (15)

EA: $C_{15}H_{12}N_4$; calc.: C, 68.56; N, 26.65; H, 4.79 meas.: C, 68.22; N, 25.76; H, 5.03



2-(4-(1,3-dineopentylimidazolidin-2-ylidene)cyclohexa-2,5-dien-1-ylidene)malononitrile (2)



Compound **9** (95 mg, 0.47 mmol, 1 eq.), dissolved in acetonitrile (5 mL), was added to a 40 °C warm solution of compound **8** (117.7 mg, 0.47 mmol, 1 eq.) in acetonitrile (7 mL). After 4 hours at 75 °C the solution was deep yellow and allowed to cool to room temperature over night. The precipitate, a yellow powder, was filtered off, washed with cooled acetonitrile (3x 3 mL) and recrystallized in acetonitrile to form neat orange crystals (113.9 mg, 0.32 mmol, 65.4%).

^{1}H NMR (500 MHz, DMSO-d_6, RT): δ (ppm) = 7.11 (d, 2H, ^{3}J = 10.0 Hz, 4), 6.86 (d, 2H, ^{3}J = 10.0 Hz, 5), 4.03 (s, 4H, 7), 3.11 (s, 4H, 8), 0.78 (s, 18H, 10)

^{13}C NMR (125 MHz, DMSO-d_6, RT): δ (ppm) =168.87 (1C, 7), 146.96 (2C, 1), 129.73 (1C, 3), 124.57 (2C, 4), 118.08 (2C, 5), 109.05 (2C, 6), 58.77 (2C, 8), 50.86 (2C, 9), 33.10 (1C, 2), 31.86 (2C, 10), 28.33 (6C, 11)

FT-IR (ATR) $\tilde{\nu}$ (cm^{-1}): 2975 (s), 2891 (m), 1648 (m), 1455 (w), 1418 (m), 1378 (m), 1318 (vw), 1085 (vs), 1045 (s), 879 (s)

UV/Vis (Ethanol) λ_{max} nm (ε [$Lmol^{-1}$ cm^{-1}]): 363 (16218)

UV/Vis (Methanol) λ_{max} nm (ε [$Lmol^{-1}$ cm^{-1}]): 354 (16800)

UV/Vis (THF) λ_{max} nm (ε [$Lmol^{-1}$ cm^{-1}]): 334 (6250), 410 (22750)

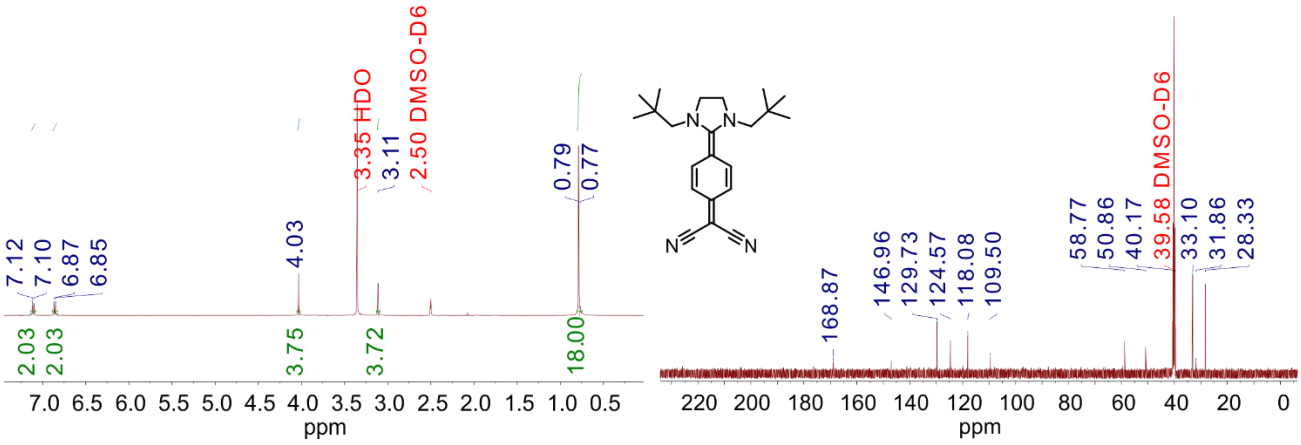
UV/Vis (ACN) λ_{max} nm (ε [$Lmol^{-1}$ cm^{-1}]): 362 (11500)

UV/Vis (DMSO) λ_{max} nm (ε [$Lmol^{-1}$ cm^{-1}]): 363 (19000)

UV/Vis (DMF) λ_{max} nm (ε [$Lmol^{-1}$ cm^{-1}]): 366 (12000)

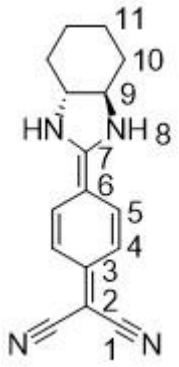
MS (ESI): m/z = 389.21 (10) [M-K^+], 373.24 (25) [M-Na^+], 351.25 (100) [M^+], 281.18 (8), 211.10 (5)

EA: $C_{22}H_{30}N_4$; calc.: C, 75.39; N, 15.98; H, 8.63 meas.: C, 75.39; N, 16.04; H, 8.77



SUPPORTING INFORMATION

2-(4-(octahydro-2H-benzo[d]imidazol-2-ylidene)cyclohexa-2,5-dien-1-ylidene)malononitrile (3)



Under argon atmosphere (1*R*,2*R*)-(−)-1,2-Diaminocyclohexane (32.2 mg, 0.28 mmol, 1 eq.), dissolved in acetonitrile (8 mL), was added to a 40 °C warm solution of **8** (70 mg, 0.28 mmol, 1 eq.) in acetonitrile (12 mL). The solution turned yellow after 2 hours at 70 °C. After cooling to room temperature, the solution stood for 2 days in the fridge. Filtration then yielded a fine grain greenish powder which was washed with cooled acetonitrile (3x 3 mL). The product was recrystallized from acetonitrile to yield fine yellow crystals (50.1 mg, 0.19 mmol, 67.2%).

^{1}H NMR (700 MHz, DMSO-d_6, RT): δ (ppm) = 10.01 (bs, 2H, 8), 7.61 (d, ^{3}J = 7.0 Hz, 2H, 4), 6.83 (d, ^{3}J = 7.0 Hz, 2H, 5), 3.43 (m, 2H, 9), 2.16 (d, 2H, 10a), 1.81 (d, 2H, 10b), 1.56 (quint., 2H, 11a), 1.36 (quint., 2H, 11b),

^{13}C NMR (176 MHz, DMSO-d_6, RT): δ (ppm) = 166.43 (1C, 7), 151.04 (1C, 1), 129.44 (1C, 3), 123.01 (2C, 4), 117.90 (2C, 5), 108.73 (2C, 6), 64.19 (2C, 9), 36.27 (1C, 2), 28.24 (2C, 10), 23.85 (2C, 11)

FT-IR (ATR) $\tilde{\nu}$ (cm^{-1}): 3182 (w), 2939 (vw), 2182 (w), 2139 (s), 1588 (vs), 1496 (vs), 1456 (m), 1331 (s), 1299 (vs), 1196 (vs), 1141 (m), 1106 (m), 832 (vs), 734 (s), 718 (m)

All Coefficients calculated from 1E-5 M Solutions

UV/Vis (Acetone) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 366 (22600), 432 (17900)

UV/Vis (DMF) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 367 (15800), 419 (31700)

UV/Vis (Ethanol) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 416 (54100)

UV/Vis (Methanol) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 407 (44000)

UV/Vis (Benzonitrile) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 438 (41500)

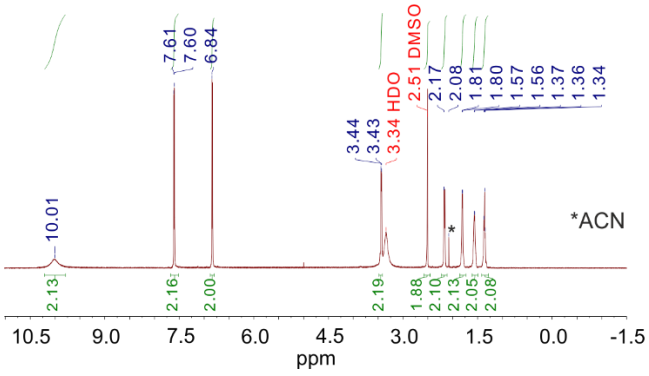
UV/Vis (THF) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 451 (46600)

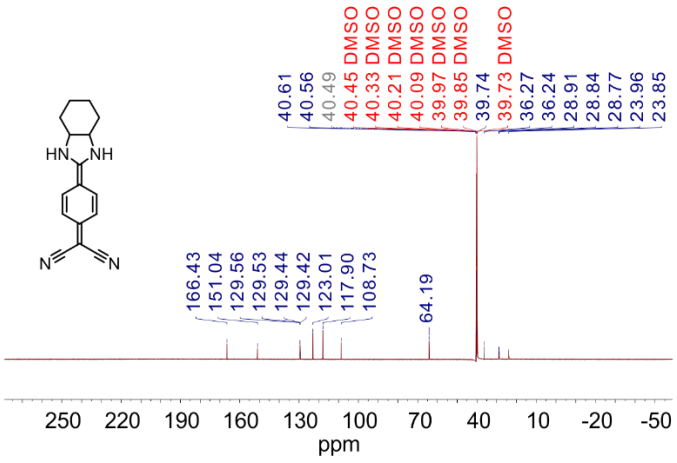
UV/Vis (DMSO) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 418 (45500)

UV/Vis (Acetonitrile) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 361 (13900), 424 (14900)

MS (EI): m/z = 264 $[MH]^{\cdot+}$ (35), 235 (20), 221 (100), 222 (25), 209 (22), 168 (15), 68 (32)

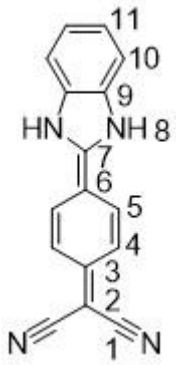
EA: $C_{16}H_{16}N_4$; calc.: C, 72.70; N, 21.20; H, 6.10; meas.: C, 72.91; N, 21.11; H, 6.12





SUPPORTING INFORMATION

2-(4-(1,3-dihydro-2H-benzo[d]imidazol-2-ylidene)cyclohexa-2,5-dien-1-ylidene)malononitrile (4)



Under argon atmosphere, ortho-Phenylendiamine (70.0 mg, 0.64 mmol, 1 eq.), dissolved in acetonitrile (15 mL), was added to a 40 °C warm solution of compound **8** (160.0 mg, 0.64 mmol, 1 eq.) in acetonitrile (20 mL). The solution turned red after 20 hours at 70 °C. After cooling to room temperature, the solution stood for 4 days in the fridge. Filtration then yielded a fine grain greenish powder which was washed with cooled acetonitrile (3x 5 mL). The product was recrystallized from acetonitrile to yield fine yellow crystals (70.2 mg, 0.27 mmol, 42.5%).

^{1}H NMR (700 MHz, DMSO-d_6, RT): δ (ppm) = 14.35 (bs, 2H, 8), 7.86 (d, ^{3}J = 8.7 Hz, 2H, 4), 7.69 (q, ^{3}J = 8.7 Hz, 2H, 11), 7.47 (q, ^{3}J = 8.6 Hz, 2H, 10), 6.95 (d, ^{3}J = 8.7 Hz, 2H, 5)

^{13}C NMR (176 MHz, DMSO-d_6, RT): δ (ppm) = 150.27 (1C, 1), 149.31 (1C, 7), 132.26 (1C, 3), 128.54 (1C, 4), 125.48 (2C, 10), 123.51 (2C, 9), 118.49 (2C, 5), 113.47 (2C, 11), 79.63 (1C, 6), 35.11 (1C, 2)

FT-IR (ATR) $\tilde{\nu}$ (cm^{-1}): 2952 (w), 2877 (w), 2849 (w), 2761 (w), 2190 (s), 2140 (s), 1637 (w), 1612 (m), 1503 (m), 1459 (m), 1387 (m), 1336 (m), 1230 (m), 1201 (m), 819 (s), 742 (s)

UV/Vis (DMF) λ_{max} nm (ε [$Lmol^{-1}$ cm^{-1}]): 391 (56500)

UV/Vis (ACN) λ_{max} nm (ε [$Lmol^{-1}$ cm^{-1}]): 385 (86800)

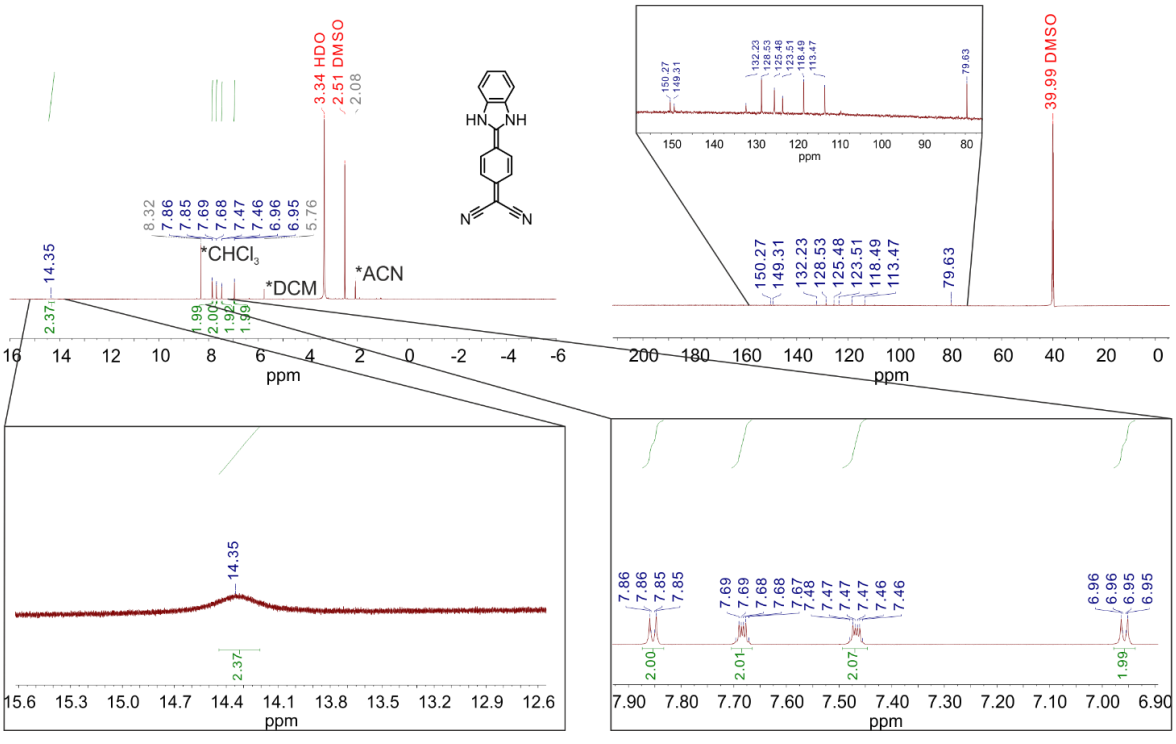
UV/Vis (DMSO) λ_{max} nm (ε [$Lmol^{-1}$ cm^{-1}]): 392 (42200)

UV/Vis (THF) λ_{max} nm (ε [$Lmol^{-1}$ cm^{-1}]): 388 (7000)

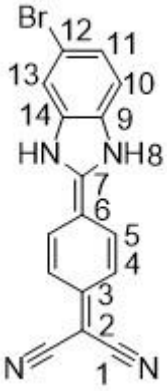
UV/Vis (MeOH) λ_{max} nm (ε [$Lmol^{-1}$ cm^{-1}]): 424 (20000)

MS (EI): m/z = 259.10 (15), 258.09 $[MH]^{\cdot+}$ (100), 257.10 (5), 232 (5)

EA: $C_{16}H_{16}N_4$; calc.: C, 74.40; N, 21.69; H, 3.90; meas.: C, 74.31; N, 21.78; H, 4.07;



2-(4-(5-bromo-1,3-dihydro-2H-benzo[d]imidazol-2-ylidene)cyclohexa-2,5-dien-1-ylidene)malononitrile (5)



Under argon atmosphere 1,2-Diamino-4-Bromobenzene (494.0 mg, 2.640 mmol, 1 eq.) was added to a solution of compound **8** (637.0 mg, 2.570 mmol, 1 eq.) in ethanol (30 mL). After 12 hours under reflux conditions the solution turned red and a yellow powder precipitated. It was filtrated and washed with cooled ethanol (3x 5 mL) to yield a yellow powder (298.7 mg, 0.885 mmol, 34.0%).

^{1}H NMR (700 MHz, DMSO-d6, RT): δ (ppm) = 14.32 (bs, 2H, 8), 7.83 (d, 1H, ^{3}J = 14.0 Hz, 4), 7.81 (s, 1H, 13), 7.58 (multiplet, 2H, ^{3}J = 14.0 Hz, 10+11), 6.93 (d, 2H, ^{3}J = 7.0 Hz, 5)

^{13}C NMR (176 MHz, DMSO-d6, RT): δ (ppm) = 151.29 (2C, 1), 149.48 (1C, 7), 134.03 (1C, 3), 131.88 (2C, 4), 128.67 (1C, 13), 128.14 (1C, 10), 123.80 (1C, 14), 118.48 (1C, 9), 117.23 (2C, 5), 116.05 (1C, 11), 115.21 (1C, 6), 109.53 (1C, 12), 35.56 (1C, 2)

FT-IR (ATR) ṽ (cm^{-1}): 819.63 (s), 856.28 (w), 1051.07 (m), 1504.28 (s), 1575.63 (w), 1604.56 (s), 2138.78 & 2192.78 (s), 2235.20 (w)

UV/Vis (DMF) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 398 (34470)

UV/Vis (ACN) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 392 (46170) (2.5E-6 mol/L Solution)

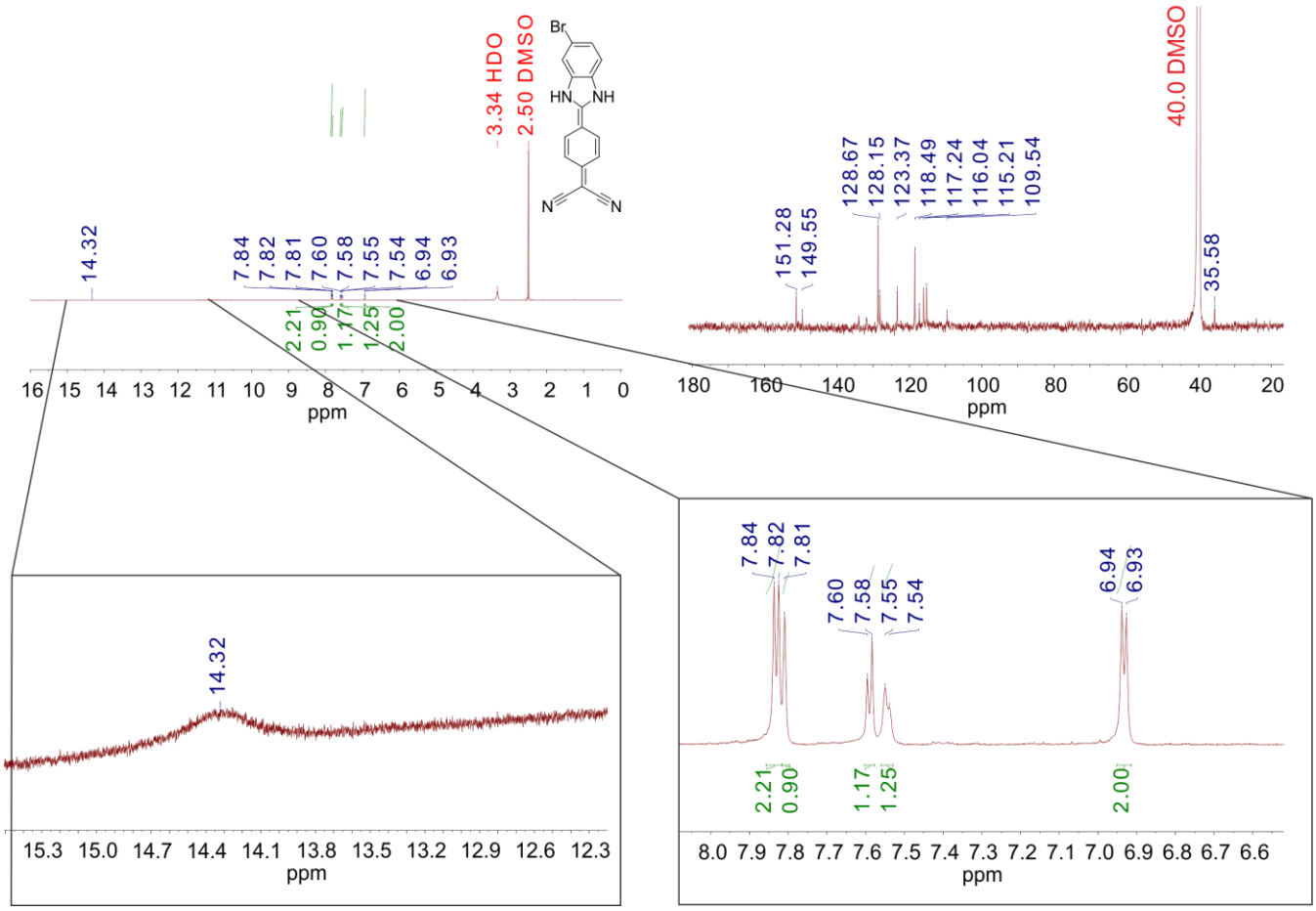
UV/Vis (DMSO) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 398 (108600)

UV/Vis (THF) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 398 (53670)

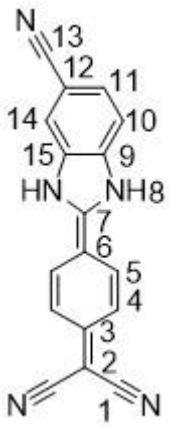
UV/Vis (MeOH) λ_{max} nm (ε [Lmol^{-1} cm^{-1}]): 317 (87490)

MS (ESI^{-}): m/z = 334.99, 335.97, 336.99, 337.99 [MH]$^{-}$ 334.99, 335.99, 336.99, 337.995

EA: $C_{16}H_9BrN_4$; calc.: C, 57.00; N, 16.62; H, 2.69; Br, 23.70; meas.: C, 57.05; N, 16.64; H, 3.61; other, 23.15;



2-(4-(5-cyano-1,3-dihydro-2H-benzo[d]imidazol-2-ylidene)cyclohexa-2,5-dien-1-ylidene)malononitrile (6)



Under argon atmosphere 3,4-Diaminobenzonitrile (39.8 mg, 0.30 mmol, 1 eq.) was added to a solution of **8** (75.0 mg, 0.30 mmol, 1 eq.) in ethanol (8 mL). After 3 days at 65 °C the solution turned red and yellow powder precipitated. It was filtrated and washed with cooled ethanol (3x 5 mL) to yield a yellow powder (16.0 mg, 0.056 mmol, 18.0%).

^{1}H NMR (700 MHz, DMSO-d6, RT): δ (ppm) = 16.00 (bs, 2H, 8), 8.18 (s, 1H, 14), 7.88 (d, ^{3}J = 7.0 Hz, 2H, 4), 7.83 (d, 1H, ^{3}J = 7.0 Hz, 11), 7.80 (d, 1H, ^{3}J = 7.0 Hz, 10), 6.95 (d, ^{3}J = 7.0 Hz, 2H, 5)

^{13}C NMR (176 MHz, DMSO-d6, RT): δ (ppm) = 153.02 (2C, 1), 149.92 (1C, 13), 136.17 (1C, 7), 133.00 (1C, 3), 128.93 (2C, 4), 128.92 (2C, 10), 123.26 (1C, 15), 119.37 (1C, 9), 118.53 (2C, 5), 118.05 (1C,14), 114.61 (1C, 11), 109.59 (1C, 6), 106.96 (1C, 12), 36.03 (1C, 2)

SUPPORTING INFORMATION

FT-IR (ATR) ṽ (cm-1): 817.67 (s), 949.77 (m), 1204.33 (m), 1340.28 (m), 1604.48 (s), 2146.38 (vs), 2196.52 (m), 2229.31 (w), 2852.2 (w)

UV/Vis (DMF) λ_{max} nm (ε [Lmol-1 cm-1]): 318 (32460)

UV/Vis (ACN) λ_{max} nm (ε [Lmol-1 cm-1]): 335 (19600)

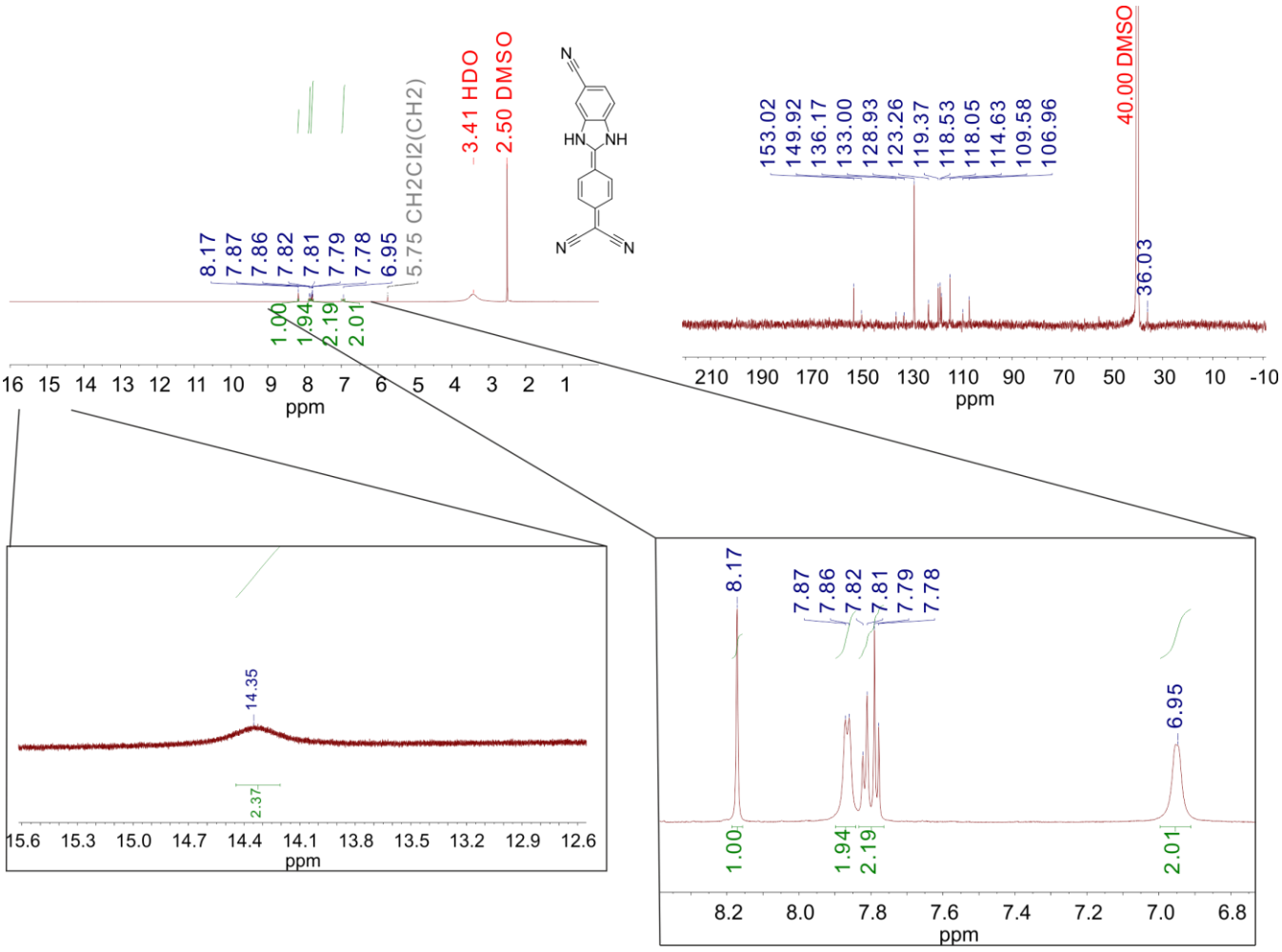
UV/Vis (DMSO) λ_{max} nm (ε [Lmol-1 cm-1]): 360 (24750)

UV/Vis (THF) λ_{max} nm (ε [Lmol-1 cm-1]): 319 (40710)

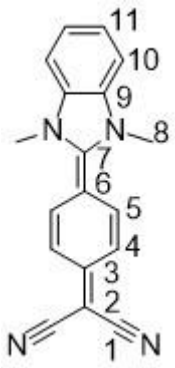
UV/Vis (MeOH) λ_{max} nm (ε [Lmol-1 cm-1]): 316 (30310)

MS (ESI$^-$): m/z = 282.07, 283.07, 284.07, 284.09 $[M]^-$ 282.079, 283.076, 283.082, 284.085

EA: $C_{20}H_{12}N_4$; calc.: C, 72.08; N, 24.72; H, 3.20; meas.: C, 67.34; N, 21.11; H, 3.23;



2-(4-(1,3-Dimethyl-1,3-dihydro-2*H*-benzo[*d*]imidazol-2-yliden)cyclohexa-2,5-dien-1-yliden)malononitrile (7)



43.5 µL *N*,*N'*-Dimethyl-1,2-phenylendiamin (0.054 g, 0.4 mmol) was added to a warm solution of **8** (0.1 g, 0.4 mmol) in acetonitrile (13 mL). The solution was stirred for 20 hours at 70 °C. After cooling to room temperature filtration yielded a fine grain yellow powder which was washed with cooled acetonitrile (3x 5 mL). Yield: 28.3 mg, 0.1 mmol, 25%.

^{1}H NMR (700 MHz, DMSO-d_6, RT): δ (ppm) = 8.01 (dd, *J* = 6.0, 3.0 Hz, 2H, 11), 7.67 (dd, *J* = 6.0, 2.9 Hz, 2H, 10), 7.49 (d, *J* = 8.5 Hz, 2H, 4), 6.98 (d, *J* = 8.2 Hz, 2H, 5), 3.92 (s, 6H, 8)

^{13}C NMR (176 MHz, DMSO-d_6, RT): δ (ppm) = 151.89 (1C, 1), 148.15 (1C, 7), 132.18 (1C, 3), 131.36 (2C, 5), 126.40 (2C, 9), 123.82 (2C, 11), 118.04 (2C, 4), 113.26 (1C, 10), 109.87 (2C, 8), 106.88 (2C, 6), 32.92 (2C, 2)

FT-IR (ATR) $\tilde{\nu}$ (cm^{-1}): 151.54 (1C, 1), 147.79 (1C, 7), 131.82 (1C, 3), 131.00 (2C, 5), 126.04 (2C, 9), 123.46 (2C, 11), 117.68 (2C, 4), 112.90 (1C, 10), 109.51 (2C, 8), 106.52 (2C, 6), 32.92 (2C, 2)

UV/Vis (DMF) λ_{max} nm (ε [$Lmol^{-1}$ cm^{-1}]): 415 (24100)

UV/Vis (ACN) λ_{max} nm (ε [$Lmol^{-1}$ cm^{-1}]): 409 (22200)

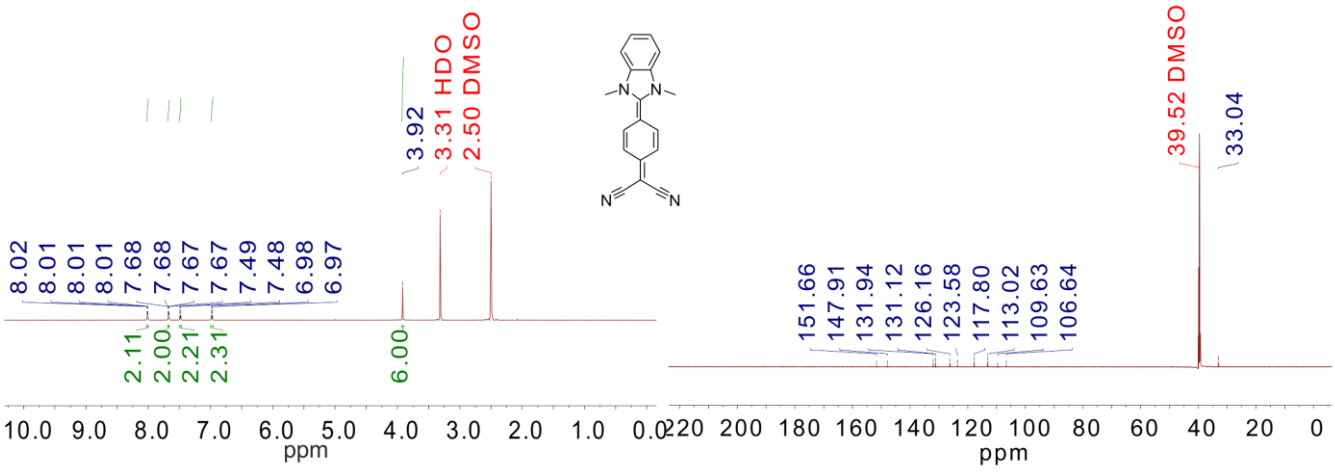
UV/Vis (DMSO) λ_{max} nm (ε [$Lmol^{-1}$ cm^{-1}]): 410 (20950)

UV/Vis (THF) λ_{max} nm (ε [$Lmol^{-1}$ cm^{-1}]): 454 (11600)

UV/Vis (MeOH) λ_{max} nm (ε [$Lmol^{-1}$ cm^{-1}]): 396 (16500)

MS (EI): m/z = 881.34 (23) [3 M-Na]$^+$, 611.20 (26) [2 M-K]$^+$, 595.22 (55) [2 M-Na]$^+$, 325.08 (90) [M-K]$^+$, 309.10 (100) [M-Na]$^+$, 287.12 (55) [MH]$^+$

EA: $C_{18}H_{14}N_4$; calc.: C, 75.50; N, 19.57; H, 4.93; meas.: C, 71.13; N, 17.76; H, 5.11;



3. Crystal Structures

*Table S1: Crystallographic data and structure refinement details of **1-3**. [a] Data taken from [1]*

	1[a]	**2**	**3**
Empirical formula	C15 H17 N5 O	C22 H30 N4	C16 H16 N4
Weight	284.34	350.50	264.33
Temperature of measurement	100 K	100 K	101 K
Crystal System	N/A	monoclinic	monoclinic
Space Group	P-1	P 21/c	$P2_1$
Unit Cell Dimensions and volume	a = 5.6328(5) b = 11.2617(10) c = 11.2686(10) α = 86.4430(10) β = 77.4390 (10) γ = 85.4860(10)	a = 9.3411(2) b = 19.6042(3) c = 11.8719(2) α = 90 β = 109.8547(7) γ = 90	a = 9.4240(2) b = 11.9409(2) c = 12.0647(2) α = 90 β = 91.446(2) γ = 90
Volume[$Å^3$]	694.81 (11)	2044.81(6)	1357.22(4)
No. of units in unit cell (Z)	2	4	4
ρ_{calc} [g/cm^{-3}]	1.354	1.139	1.294
μ [mm^{-1}]	0.090	0.068	0.605
F(000)	300.0	760.0	560.0
Crystal size [mm^{-3}]	0.5 x 0.34 x 0.1	0.560 × 0.480 × 0.180	0.370 x 0.060 x 0.020
Radiation and wavelength	MoKα (λ = 0.71073)	MoKα (λ = 0.71073)	MoKα (λ = 1.54178)
2θ range for data collection [°]	3.7 to 52.76	4.636 to 51.464	7.33 to 133.602
Index ranges	-7 ≤ h ≤ 7, -14 ≤ k ≤ 14, -14 ≤ l ≤ 14	-11 ≤ h ≤ 11, -23 ≤ k ≤ 23, -14 ≤ l ≤ 14	-11 ≤ h ≤ 11, -12 ≤ k ≤ 14, -14 ≤ l ≤ 14
No. of Refl	7425	19722	11121
Independent Refl.	2813 [R_{int} = 0.0360, R_{sigma} = N/A]	3889 [R_{int} = 0.0310, R_{sigma} = 0.0228]	4200 [R_{int} = 0.1331, R_{sigma} = 0.1230]
Data/restraints/parameters	2813/0/258	3889/0/242	4200/1/356
Goodness-of-fit on F^2	1.047	1.037	1.032
Final R indexes [I>=2σ (I)]	R_1 = 0.0360, wR_2 = 0.1000	R_1 = 0.0384, wR_2 = 0.0918	R_1 = 0.0704, wR_2 = 0.1415
Final R indexes [all data]	R_1 = 0.0375, wR_2 = 0.1016	R_1 = 0.0444, wR_2 = 0.0954	R_1 = 0.1221, wR_2 = 0.1636
Largest diff. peak/hole [e $Å^{-3}$]	0.24/-0.23	0.30/-0.21	0.34/-0.27

Despite multiple efforts to crystalize compound **4** from different solvents and solvents mixtures it was not possible to get a measurable single crystal of the molecule.

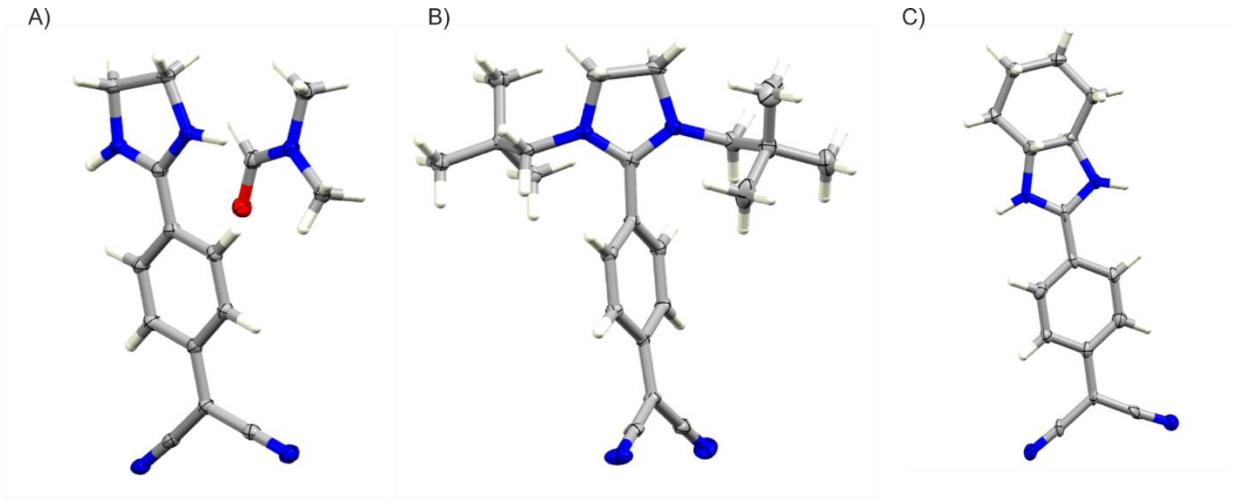


*Figure S1: Crystal structures of compound **1** (A), taken from [1] crystallized with one molecule of dimethylformamide per unit cell, compound **2** (B) and compound **3** (C).*

4. UV/Vis and Fluorescence measurements

The absorption and fluorescence spectra are recorded at concentrations of 10^{-6} M to avoid association phenomena.

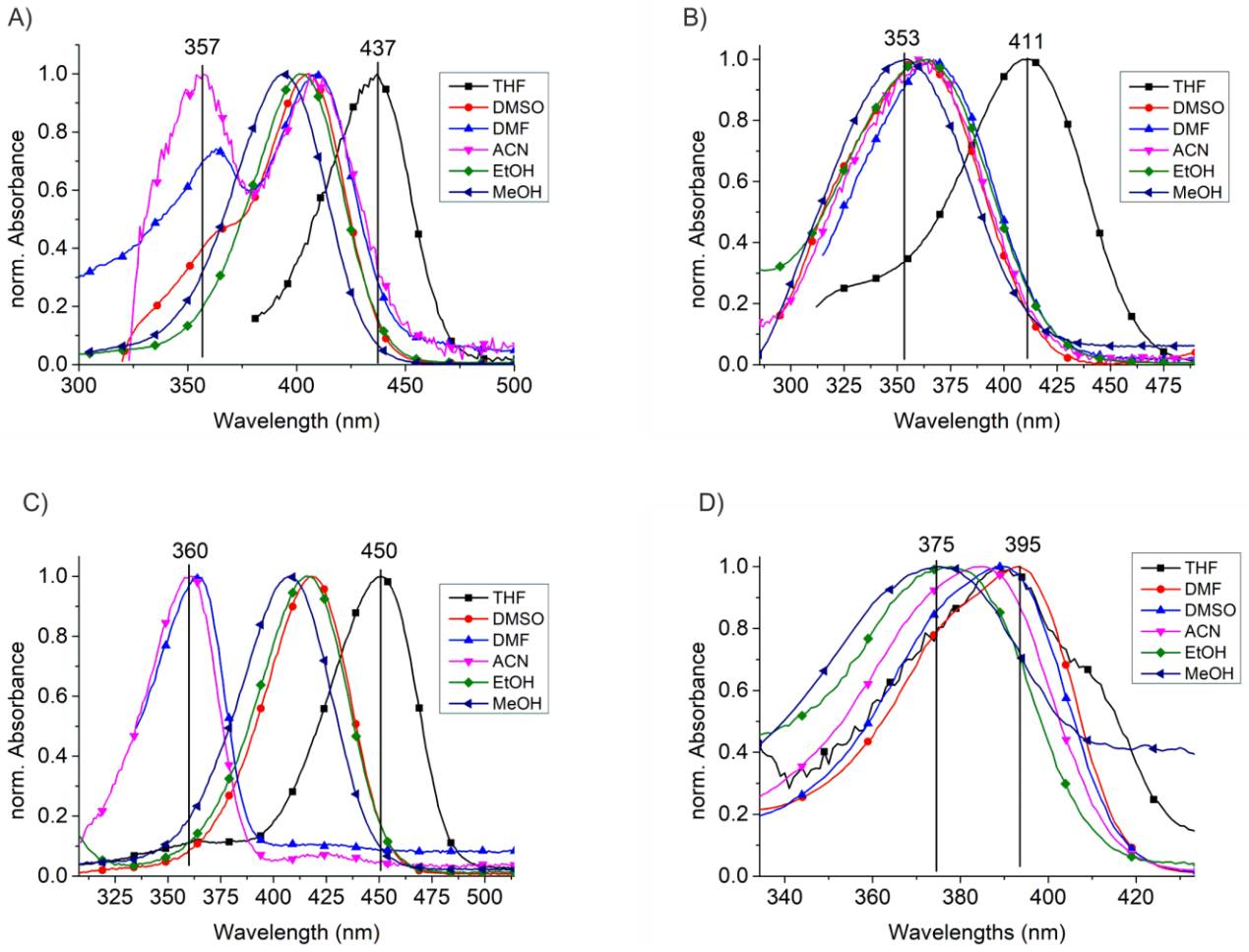


*Figure S2: Normalized absorption spectra of compounds A) ethyl-bridged (**1**) B) Neopentyl-ethyl-bridged (**2**) C) cyclohexyl-bridged (**3**) D) benzene-bridged (**4**) in six solvents of increasing polarity: tetrahydrofuran, dimethylformamide, dimethyl sulfoxide, acetonitrile, ethanol, methanol.*

Table S2: Absorption- and fluorescence-maxima with calculated stokes shift for compound 1-4

		1			**2**			**3**			**4**		
E_T^N	Solvent	Abs_{max} [nm]	Em_{max} [nm]	Stokes Shift [nm]	Abs_{max} [nm]	Em_{max} [nm]	Stokes Shift [nm]	Abs_{max} [nm]	Em_{max} [nm]	Stokes Shift [nm]	Abs_{max} [nm]	Em_{max} [nm]	Stokes Shift [nm]
0.207	tetrahydrofuran	436	478	42	410	489	79	450	490	40	389	431	42
0.386	dimethylformamide	363	405	42	366	490	124	364	408	44	393	430	37
0.444	dimethyl sulfoxide	359	435	76	362	441	79	418	465	47	389	436	47
0.460	acetonitrile	355	417	62	361	500	139	360	413	53	385	439	54
0.654	ethanol	402	456	54	362	450	88	417	462	45	314	404	90
0.762	methanol	394	450	56	354	486	132	408	461	53	425	486	61

5. Quantum Yield and Fluorescence Lifetimes measurements

Photoluminescence quantum yields (Φ) were determined absolutely with an integrating sphere setup from Hamamatsu (Quantaurus-QY C11347-11). All Φ measurements were performed at 25 °C using special 10 mm x 10 mm long neck quartz cuvettes from Hamamatsu. Values below 1% quantum yield are not reliable in the measurement setup and are therefore given as < 1%. The uncertainty of Φ measurements over 10% absolute value is about 5%.[2]

The fluorescence lifetime (t), the average time in which the fluorophore is in an excited state before it relaxes to the ground state, was recorded on a fluorometer FLS 920 (Edinburgh Instruments) equipped with a Hamamatsu R3809U-50 (range 200–850 nm, response width <25 ps), Multi-Channel Plate (MCP) detector, Czerny-Turner double monochromators and either a supercontinuum laser (Fianium SC400-2-PP) or a Edinburgh Instrument EPLED-375 (picosecond pulsed light emitting diode) for excitation at 375 nm, or a Edinburgh Instrument EPL-330 (picosecond pulsed diode laser) for excitation at 330 nm. All the measurements were performed at T = 298 K using 10 mm x 10 mm quartz cuvettes from Hellma GmbH always filled with 2 mL of solvent or dye solution. Before each measurement, the instrument response function (IRF) was measured, which is important for the correct fit of the then measured data. The lifetime measurements were analysed with Edinburgh Instruments FAST Software and fitted with a reconvolution fit. All the lifetimes could be evaluated mono, bi- or tri-exponentially with a reduced X^2 between 0.8 and 3.0. An exemplary formular for a multi exponential decay is given in Equations S1. From the multi-exponential decays, the intensity-weighted average lifetimes were calculated using Equation S2 and the fit parameter given by the FAST Software (see Figure S6 and Figure S8 / Figure S9)

$$\mathrm{I}\,(\mathrm{t}) = I_0 \times \sum \mathrm{B_i}\, e^{\frac{-t}{\tau_i}}$$

Equation S1: Formula for a multi-exponential decay. B_i is the pre-exponential factor. B_i1 is the amplitude of the first exponential term, B_i2 that of the second one and so on. τ_i is the fluorescence lifetime.

$$\tau = \frac{\sum \mathrm{B_i}\, \tau_i^{\,2}}{\sum \mathrm{B_i}\, \tau_i}$$

Equation S2: Equation used for the evaluation of the fit parameters B_i and τ_i given by by the reconvolution fit of the multi-exponential decays to the measured fluorescence decays (see Figure S8).

❖ **Exponential Components Analysis (Reconvolution)**

Fitting range [679; 2000] channels

χ^2 1.067

	B_i	ΔB_i	f_i (%)	Δf_i (%)	τ_i (ns)	$\Delta\tau_i$ (ns)
1	0.0211	0.0014	2.616	2.67	0.271	0.259
2	0.1213	0.0008	97.384	0.668	1.756	0.0007

Shift	0.012	(±	0.926	ns)
Decay Bac	2.655	(±	0.198	)
IRF backgr	0			

Figure S3: Exemplary fit parameters given by the reconvolution fit of the multi-exponential decays using the Edinburgh Instruments FAST Software. The fit parameters B_i and τ_i are used according to equation S2 to calculate the average lifetime of the excited state. B_i is the pre-exponential factor. ΔB_i Is the standard deviation of the associated B value. f_i is the "fraction of fluorescence intensity" in %. It is calculated from B_i and τ_i. f_i characterises the brightness of the fluorescence component (which scales with the pre-exponential factor and the lifetime). Δf_i is the associated error of f_i. $\Delta\tau_i$ standard deviation of the calculated lifetime. A shift parameter is fitted alongside the lifetime parameters. This shift characterises the delay of the decay process in respect to the IRF measurement. The fluorescence decay might be superimposed by a constant background. This background often originates from dark count rates of the detector, but might also be caused by light leaks to ambient light. IRF Background is the analogue to the decay background. Both should be close to zero. A constant background of hundreds or even thousands of counts in average is not unusual.

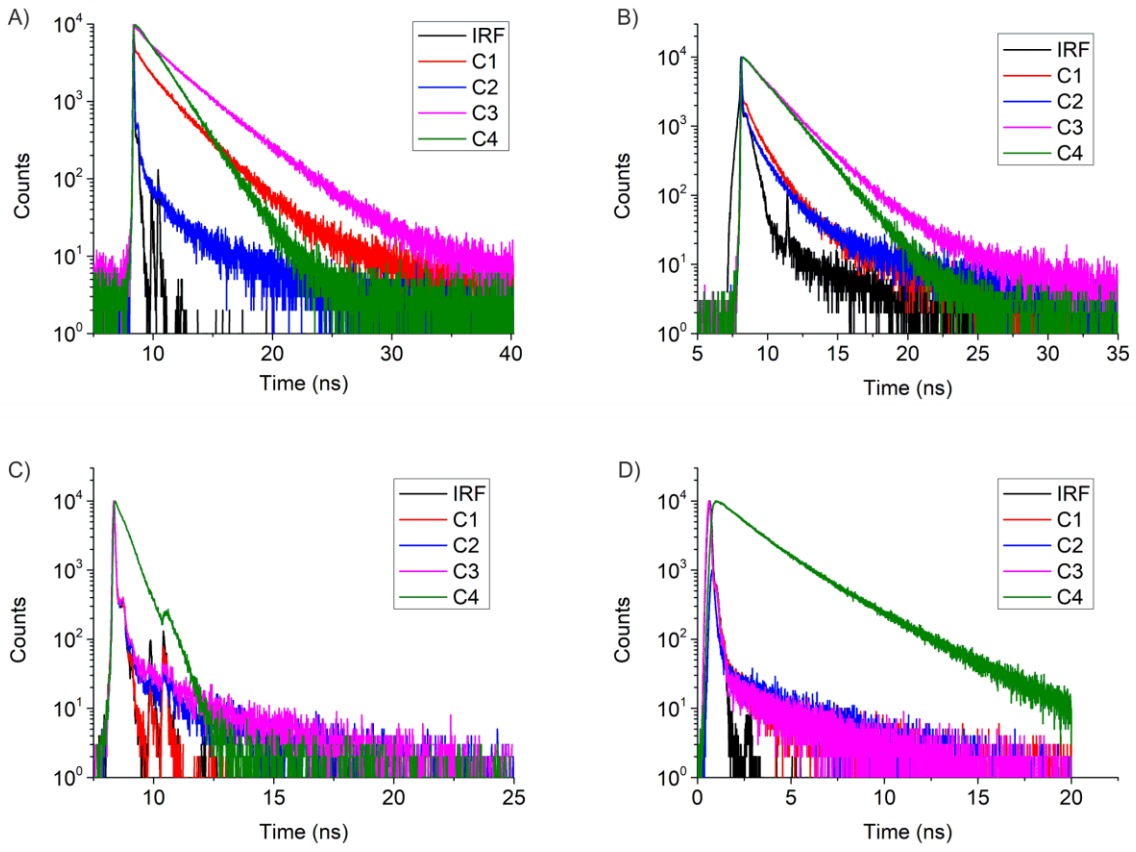


*Figure S4: Fluorescence lifetime decays of compound **1-4** (**1** red, **2** blue, **3** pink, **4** green) in dimethylformamide (A), DMSO (B), methanol (C) and tetrahydrofuran (D).*

A)

v Exponential Components Analysis (Reconvolution)						
Fitting range	[695; 2000] channels					
χ^2	1.122					
	B_i	DB_i	f_i (%)	Df_i (%)	t_i (ns)	Dt_i (ns)
1	0.1315	0.0005	99.486	0.535	1.815	0.002
2	0.0001	0.0003	0.514	1.252	9.323	0.378
Shift	-0.046 (±	0.576	ns)			
Decay Background	0.707 (±	7.69	)			
IRF background	0.2					

B)

v Exponential Components Analysis (Reconvolution)						
Fitting range	[679; 2000] channels					
χ^2	1.067					
	B_i	DB_i	f_i (%)	Df_i (%)	t_i (ns)	Dt_i (ns)
1	0.0211	0.0014	2.616	2.67	0.271	0.259
2	0.1213	0.0008	97.384	0.668	1.756	0.0007
Shift	0.012 (±	0.926	ns)			
Decay Background	2.655 (±	0.198	)			
IRF background	0					

C)

v Exponential Components Analysis (Reconvolution)						
Fitting range	[689; 2000] channels					
χ^2	0.922					
	B_i	DB_i	f_i (%)	Df_i (%)	t_i (ns)	Dt_i (ns)
1	0.3521	0.1596	8.321	9414.047	0.015	17.19
2	0.0786	0.2477	51.033	690.807	0.417	4.335
3	0.0625	0.2477	40.646	693.117	0.418	5.474
Shift	0.017 (±	0.3	ns)			
Decay Background	0.987 (±	0.073	)			
IRF background	0.2					

*Figure S5: Exemplary Fit parameters given by the reconvolution fit of the multi-exponential decays using the Edinburgh Instruments FAST Software for compound **4** in A) DMF, B) DMSO and C) MeOH. The decays in DMF and DMSO were fitted with a two-exponential function, the decay in MeOH with a three-exponential function. D denotes Δ (see Figure S3 for details on all given values).*

SUPPORTING INFORMATION

A)

v Exponential Components Analysis (Reconvolution)							
Fitting range	[261; 4000] channels						
χ^2	1.313						
	B_i	DB_i	f_i (%)	Df_i (%)	t_i (ns)	Dt_i (ns)	
1	0.0112	9.10E-06	100	0.099	1.82	0.0003	
Shift	0.001 (±	0.024	ns)				
Decay Background	1.247 (±	0.114	)				
IRF background	920						

B)

v Exponential Components Analysis (Reconvolution)							
Fitting range	[200; 2500] channels						
χ^2	1.109						
	B_i	DB_i	f_i (%)	Df_i (%)	t_i (ns)	Dt_i (ns)	
1	0.0152	0.0008	59.857	3.552	1.67	0.014	
2	0.0049	0.0008	40.143	6.539	3.494	0.018	
Shift	-0.084 (±	1.168	ns)				
Decay Background	-11.561 (±	8.734	)				
IRF background	5.9						

C)

v Exponential Components Analysis (Reconvolution)							
Fitting range	[609; 4000] channels						
χ^2	1.078						
	B_i	DB_i	f_i (%)	Df_i (%)	t_i (ns)	Dt_i (ns)	
1	0.0856	0.0016	5.295	3.882	0.176	0.125	
2	0.0595	0.0003	38.47	0.241	1.838	0.003	
3	0.016	0.0001	56.236	0.376	9.992	0.0005	
Shift	-0.024 (±	0.432	ns)				
Decay Background	15.33 (±	0.69	)				
IRF background	0.2						

*Figure S6: Fit parameters given by the reconvolution fit of the multi-exponential decays using the Edinburgh Instruments FAST Software for compound **4** in A) EtOH B) THF and C) CAN. The decays in DMF and DMSO were fitted with a two-exponential function, the decay in MeOH with a three-exponential function. D denotes Δ (see Figure S3 for details on all given values).*

*Table S3: Fluorescence Quantum Yields, Fluorescence lifetimes and calculated radiative / non-radiative rates of Compounds **1-4** in Solvents of different Polarity. Values for the normalized Dimroth-Reichardt parameter E_T^N were taken from ref.[3]*

		1		2		3		4	
E_T^N	Solvent	Φ [%]	τ [ns]	Φ [%]	τ [ns]	Φ [%]	τ [ns]	Φ [%]	τ [ns]
0.207	THF	7.0	< 0.2	< 1	0.42	7.1	< 0.2	73.0	2.40
0.386	DMF	6.8	2.14	< 1	< 0.2	9.8	3.06	71.9	1.84
0.444	DMSO	10.1	0.75	< 1	0.43	7.1	2.79	92.4	1.79
0.460	ACN	7.3	< 0.2	< 1	2.40	12.2	6.74	71.3	6.33
0.654	EtOH	< 1	< 0.2	< 1	< 0.2	< 1	< 0.2	55.7	1.82
0.762	MeOH	< 1	< 0.2	< 1	3.18	< 1	< 0.2	34.9	0.38

Values below 0.2 ns are not reliable in this measurement setup and are therefore given as "< 0.2" ns.

Radiative and non-radiative rates were calculated using the following formula[4]:

$$k_r = \frac{\phi}{\tau} \qquad k_{nr} = \frac{1}{\tau} - k_r$$

Equation S3: Relationship between radiative rate, quantum yield Φ and fluorescence lifetime τ (left side).
Equation S4: Relationship between fluorescence lifetime τ, radiative rate k_r and non-radiative rate k_{nr}.

*Table S4: Radiative and non-radiative decay rates calculated from measured quantum yields and fluorescence lifetimes of compound **1-4**. Values where determined from the equations given in reference [5] a) (10^8 s^{-1}). Values are omitted where the measurement of either quantum yield or fluorescence lifetime were below the measurement limit.*

		1		2		3		4	
E_T^N	Solvent	k_r (10^8 s^{-1})	k_{nr} (10^8 s^{-1})	k_r (10^8 s^{-1})	k_{nr} (10^8 s^{-1})	k_r (10^8 s^{-1})	k_{nr} (10^8 s^{-1})	k_r (10^8 s^{-1})	k_{nr} (10^8 s^{-1})
0.207	THF	--	--	--	--	--	--	3.04	1.13
0.386	DMF	0.32	4.36	--	--	0.32	2.95	3.19	1.53
0.444	DMSO	1.35	0.12	0.05	23.21	0.25	3.33	5.16	0.43
0.460	ACN	--	--	--	--	0.18	1.30	1.13	0.45
0.654	EtOH	--	--	--	--	--	--	3.06	2.43
0.762	MeOH	--	--	--	--	--	--	9.18	17.1

6. Fluorescence measurement in solvent mixtures of differing viscosity and polarity

To study the fluorescence behaviour in solvents of different viscosity ethanol-poly ethylene glycol mixtures of the same concentration were prepared and measured at constant parameters, i.e. slid width, excitation wavelengths etc., for each molecule respectively. From the absolute determined quantum yield in pure ethanol, we have then calculated the relative quantum yields in the EtOH-PEG mixtures by literature known procedures. [5] Solvent viscosities for all solvents used are listed in table S5. Since the difference in solvent viscosity is small and dimethylformamide, as one example, shows relatively high quantum yields for all molecules, but has a lower viscosity than dimethyl sulfoxide and methanol, the fluctuation of solvent viscosity without addition of an extremely viscous solvent like PEG can be neglected. With addition of 50% PEG to ethanol, the viscosity increases to 8.3 mPa s, in a mixture of 80% PEG the viscosity is 34.1 mPa s.

Table S5: Viscosity of the used solvents, ordered in increasing polarity. [a] Values taken from reference [6]

E_T^N	Solvent	Viscosity [25 °C, mPa s][a]
0.207	tetrahydrofuran	0.55
0.386	dimethylformamide	0.82
0.444	dimethyl sulfoxide	2.00
0.460	acetonitrile	0.37
0.654	ethanol	1.08
0.762	methanol	0.60
--	PEG400/EtOH (80/20)	34.1

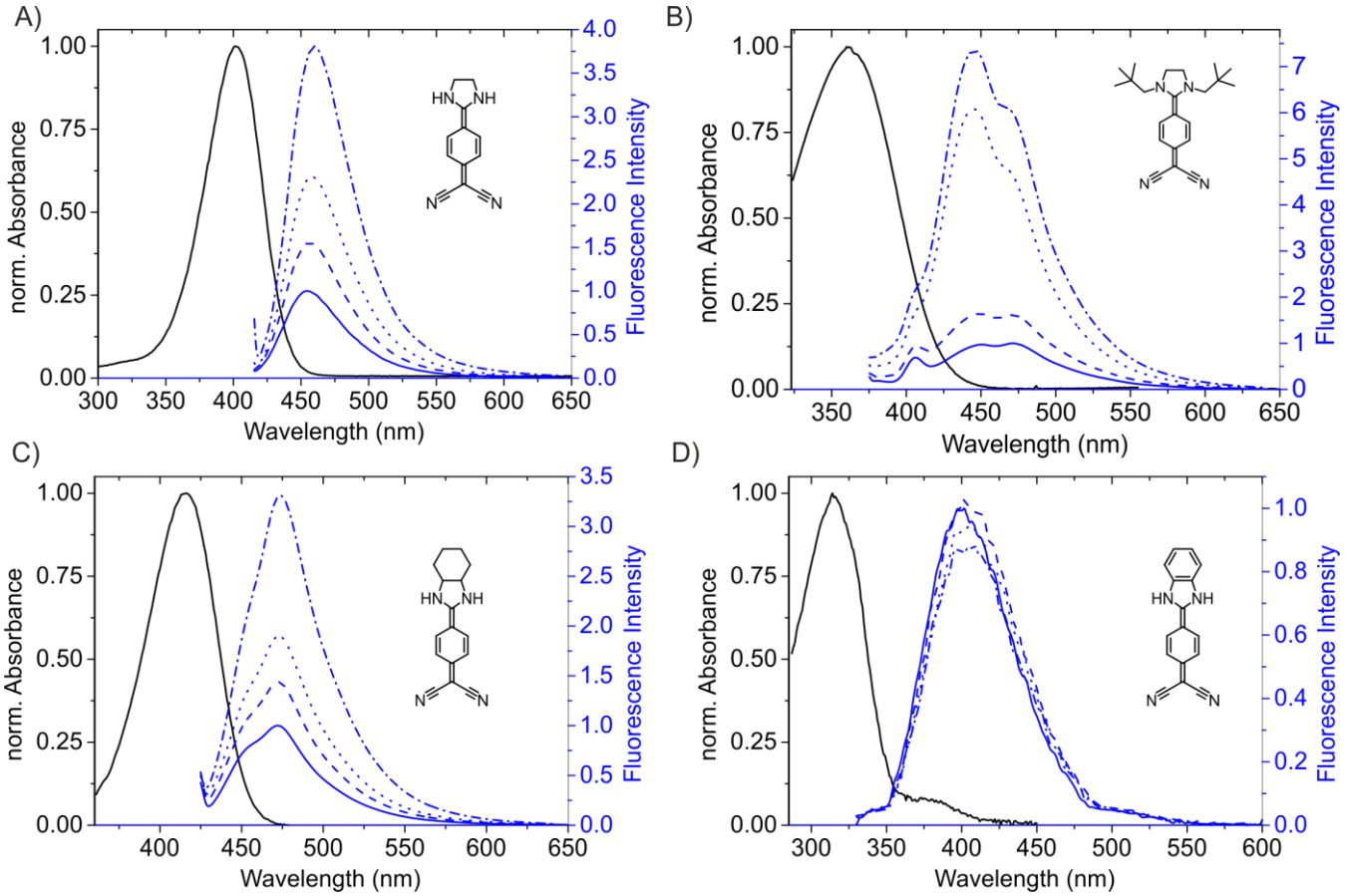


*Figure S7: Normalized absorption spectrum and four emission spectra, respectively, of EtOH-PEG mixtures ranging from 0 to 75% PEG. A) Compound **1** Exc. Wavelength: 400 nm; Slid width 5 nm, Fluorescence Increase from 0% to 75% PEG by factor 3.8 B) Compound **2**, Exc. Wavelength: 360 nm; Slid width 10 nm, Fluorescence Increase from 0% to 75% PEG by factor 7.6 C) Compound **3**, Exc. Wavelength: 415 nm; Slid width 5 nm, Fluorescence Increase from 0% to 75% PEG by factor 3.3 D) Compound **4**, Exc. Wavelength: 315 nm; Slid width 3 nm, Fluorescence Increase from 0% to 25% PEG by factor 1.03, decrease from 0% to 50 and 75% PEG.*

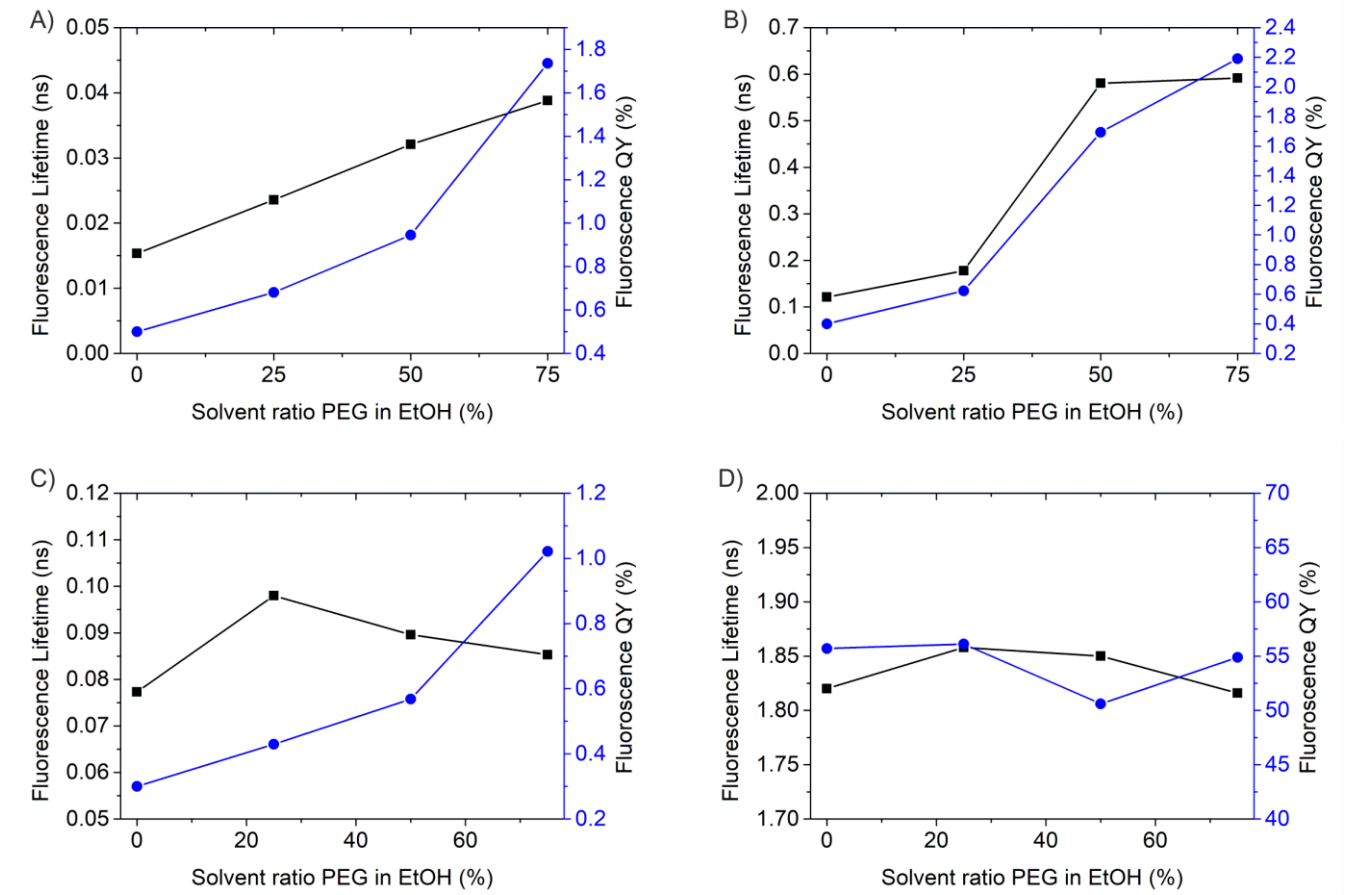


*Figure S8: Fluorescence Lifetime (square) and Fluorescence Quantum Yield (cycle) of EtOH-PEG mixtures ranging from 0 to 75% PEG for compound **1-4** (A-D).*

7. Fluorescence Measurements at elevated and reduced Temperatures

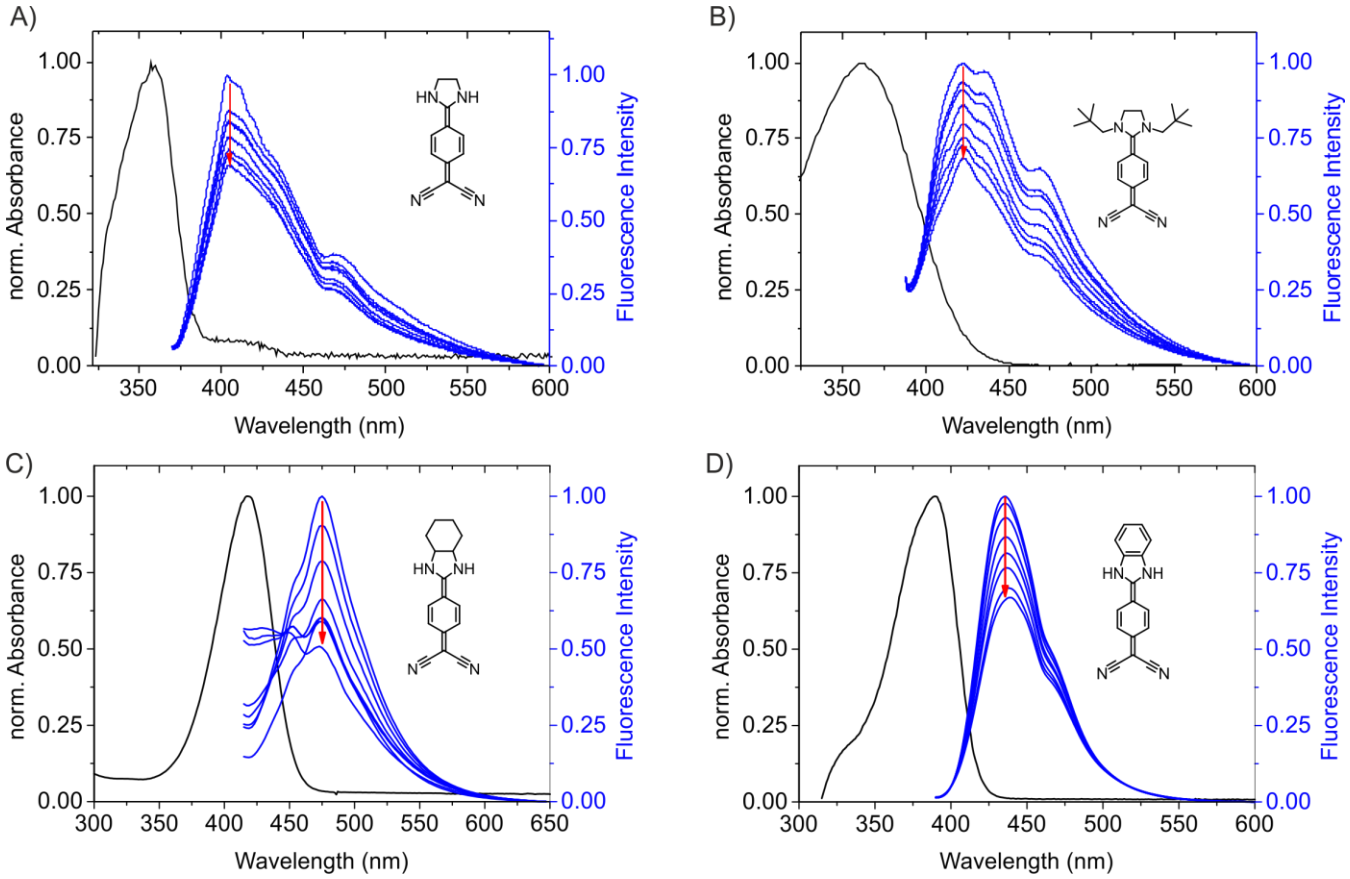


*Figure S9: normalized Absorption spectrum (black) and fluorescence spectra (blue) at increasing temperature (red arrow, 40 to 110 °C in 10 °C steps) of **1-4** (A-D) in DMSO. The fluorescence decreases from 40 °C to 110 °C by 31% per for compound **1**, 32% per for compound **2**, 47% per for compound **3** and 33% for compound **4**.*

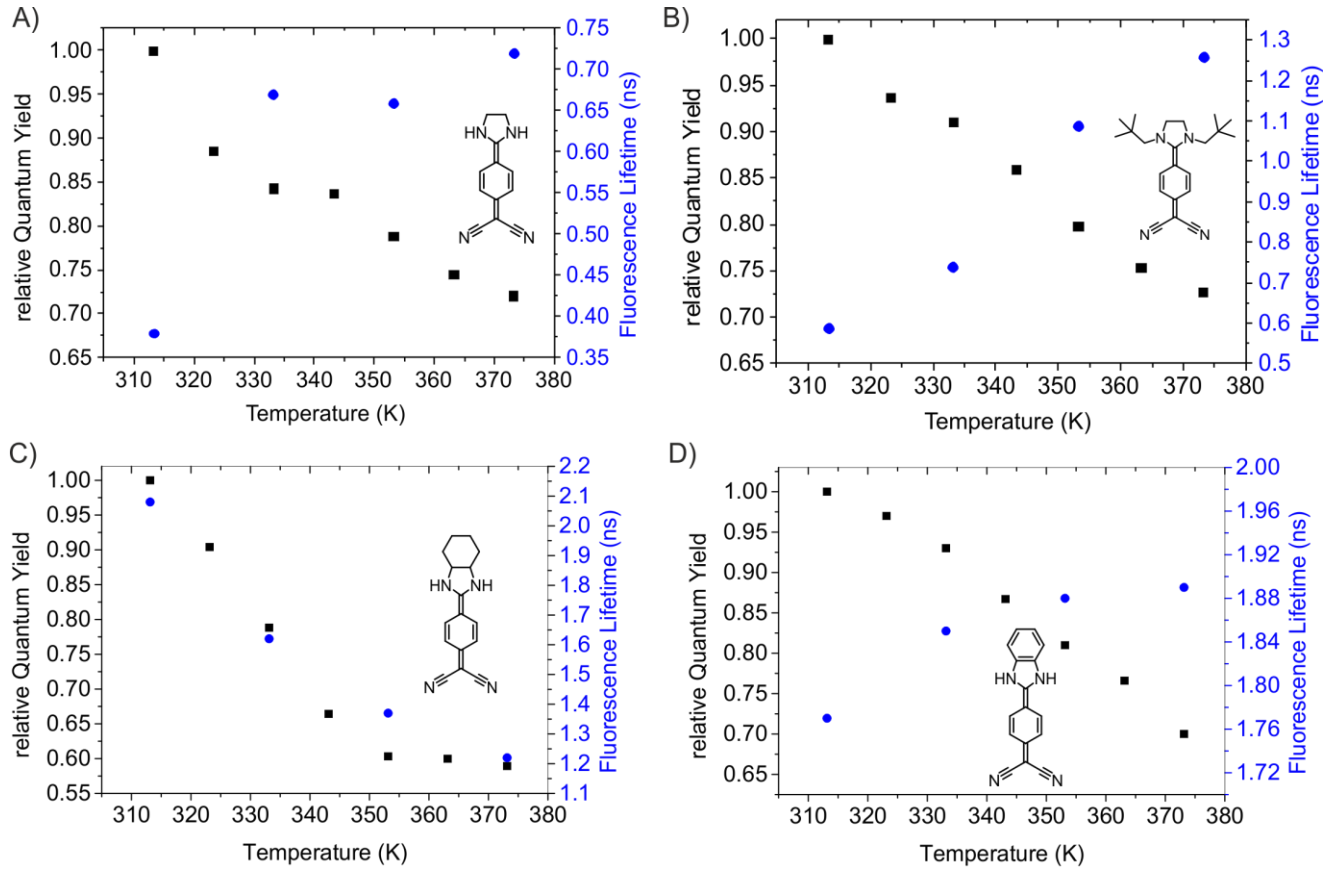


*Figure S10: Relative Quantum Yields (normalized to the spectra at 40 °C) and Fluorescence lifetimes of Compound **1-4** (A-D) upon heating in DMSO in 10 °C (QY) and 20 °C (Fluorescence lifetime) steps. Whereas the QY decreases for all compounds, the lifetime increases for **1-2** and **4** but decreases for **3**. The change of fluorescence lifetime is relatively small for compound **4** (1.77 ns (40 °C) to 1.89 ns (100 °C)) compared to **1-3**. For example, compound **2** changes from 0.59 ns (40 °C) to 1.26 (100 °C), more than 100% change.*

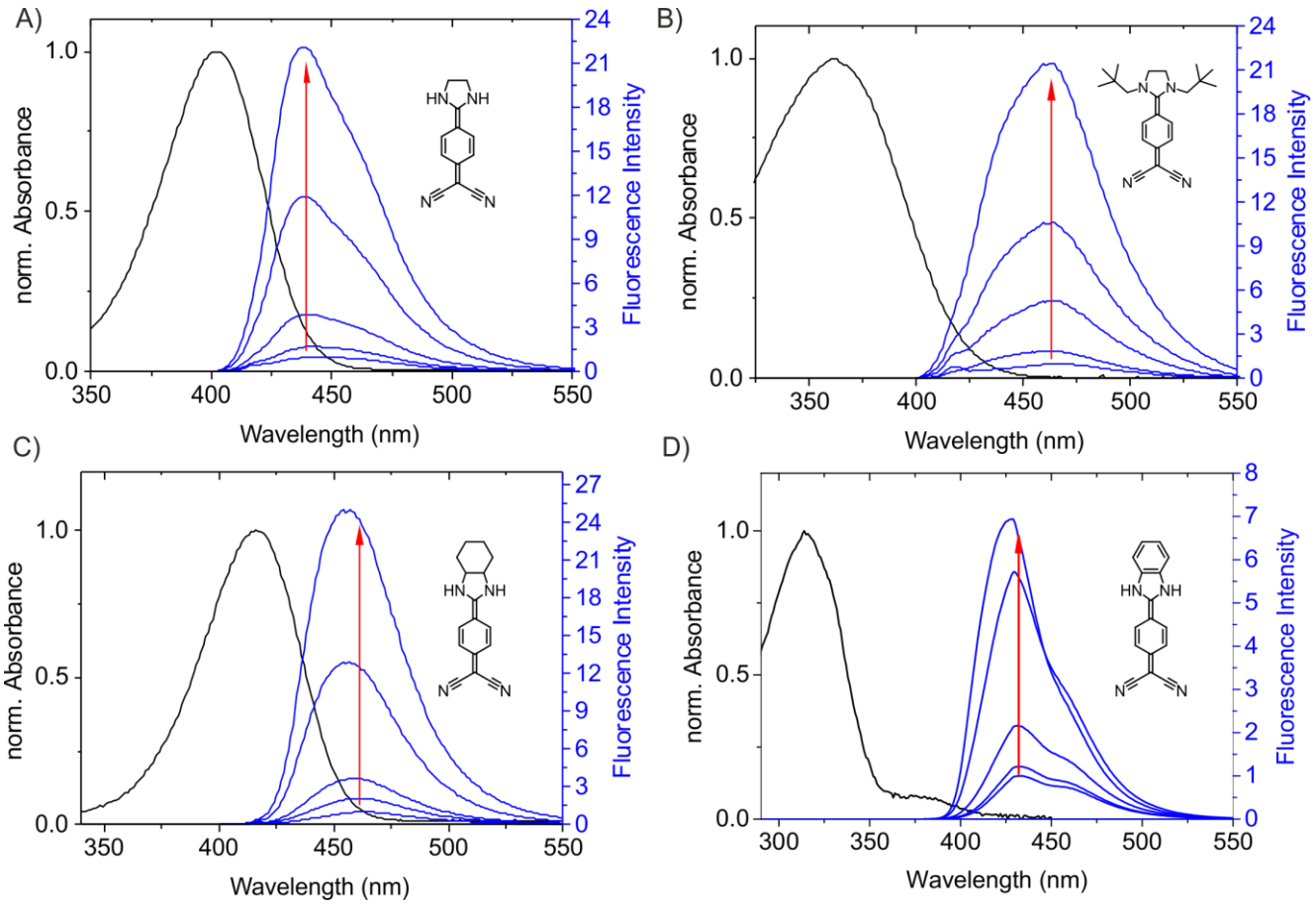


*Figure S11: normalized Absorption spectra (black) and fluorescence spectra at decreasing temperature (red arrow, from 0 °C to -100 °C in 25 °C steps) of compound **1-4** (A-D) in EtOH. The fluorescence was normalized to the value at 0 °C respectively. The Fluorescence increases from 0 °C to -100 °C about 7 times for compound **4** (D), and over 20 times for compound **1-3** (25 for compound **3**)*

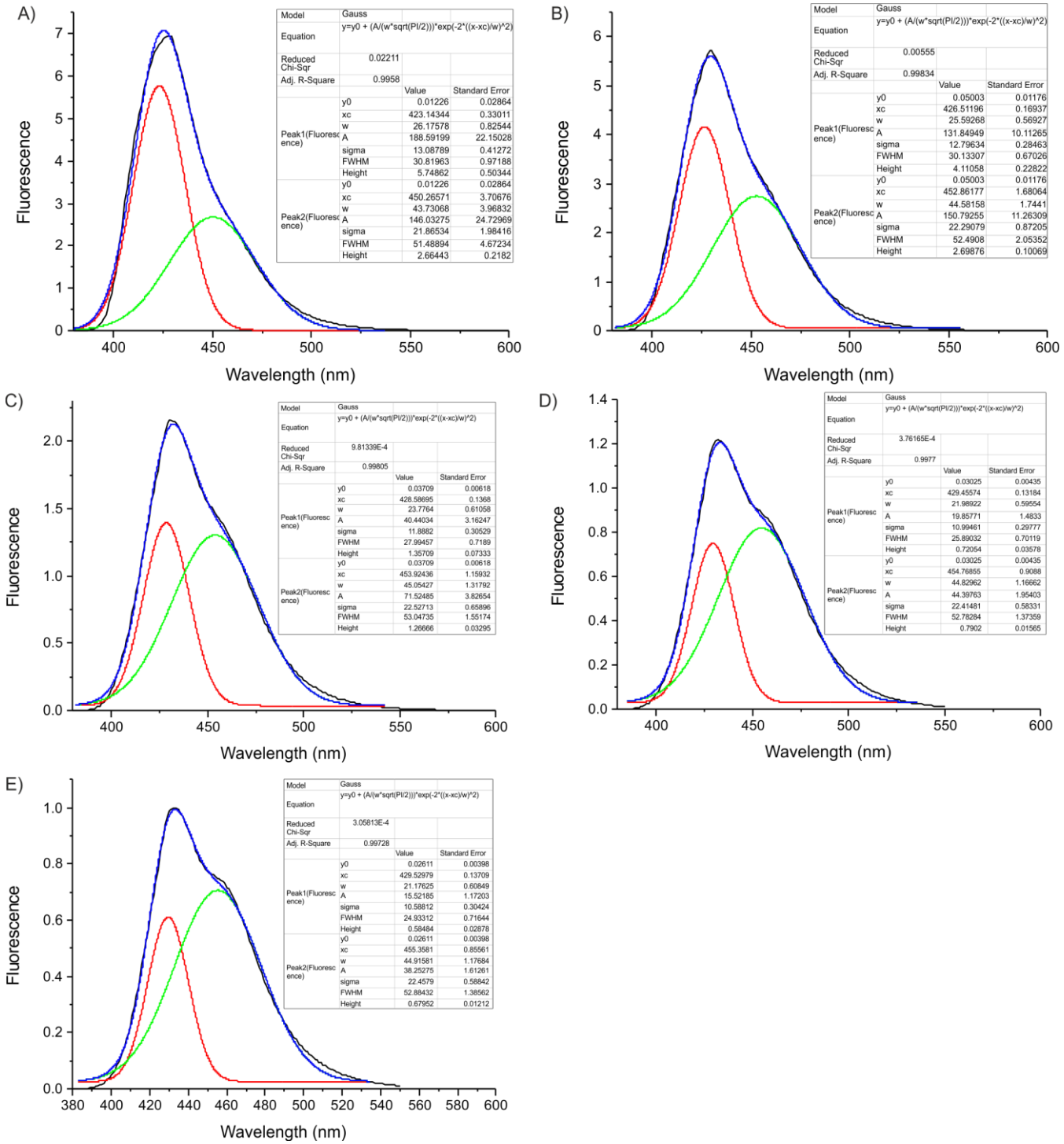


*Figure S12: Fluorescence spectra (black) and fitted spectra (blue) of compound **4** upon cooling in EtOH (depicted in Figure S11D) A) 173 K B) 198 K C) 223 K D) 248 K E) 273 K. Red and green are the two peaks fitted inside the main peak, indicating that two species might be responsible for the emission. The ratio of the peaks change with temperature.*

8. Comparison of benzene functionalized derivatives

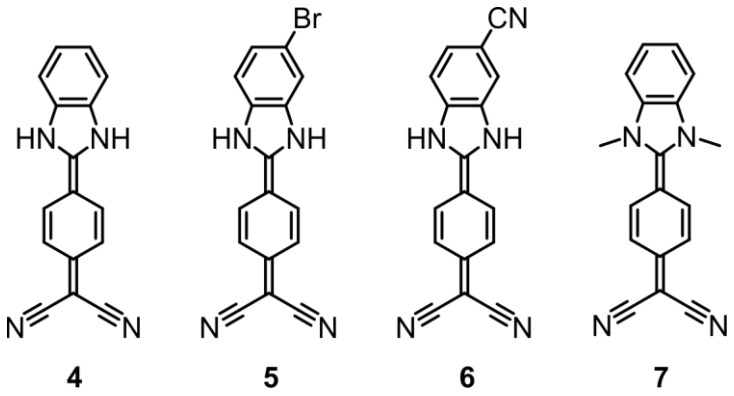


Figure S13: Molecular structures of compound 4-7

*Table S6: Fluorescence Quantum Yields and Fluorescence lifetimes of Compounds **4-7** in Solvents of different Polarity. Values for the normalized Dimroth-Reichardt parameter E_T^N were taken from ref. [3]*

		4		5		6		7	
E_T^N	Solvent	Φ [%]	τ [ns]	Φ [%]	τ [ns]	Φ [%]	τ [ns]	Φ [%]	τ [ns]
0.207	THF	73.0	2.4	33.9	0.5	68.5	1.4	< 1	--
0.386	DMF	71.9	1.8	61.6	1.8	81.3	2.0	3.3	0.5
0.444	DMSO	92.4	1.8	38.5	1.4	90.1	2.1	< 1	0.5
0.460	ACN	71.3	6.3	17.9	2.0	53.1	2.1	< 1	1.1
0.654	EtOH	55.7	1.8	62.8	1.4	70.3	1.8	< 1	--
0.762	MeOH	34.9	0.4	19.8	0.5	30.5	1.4	< 1	< 0.2

9. Cyclic voltammetry measurements

Cyclic voltammetry was carried out in dry and degassed dimethylformamide (DMF) solutions with tetrabutylammonium hexafluorophosphate (0.1 M) using a three-electrode configuration (glassy carbon, Pt counter electrode, Ag wire as pseudoreference) and a PAR VersaSTAT 4 (Princeton Applied Research) potentiostat. The S3 decamethylferrocene/decamethylferrocenium ([FeCp2*] $^+$/0) was used as the internal reference and referred to the ferrocene/ferrocenium (FcH^+/0) couple.[7] The second and third cycle are always depicted and were taken for all evaluation of the data.

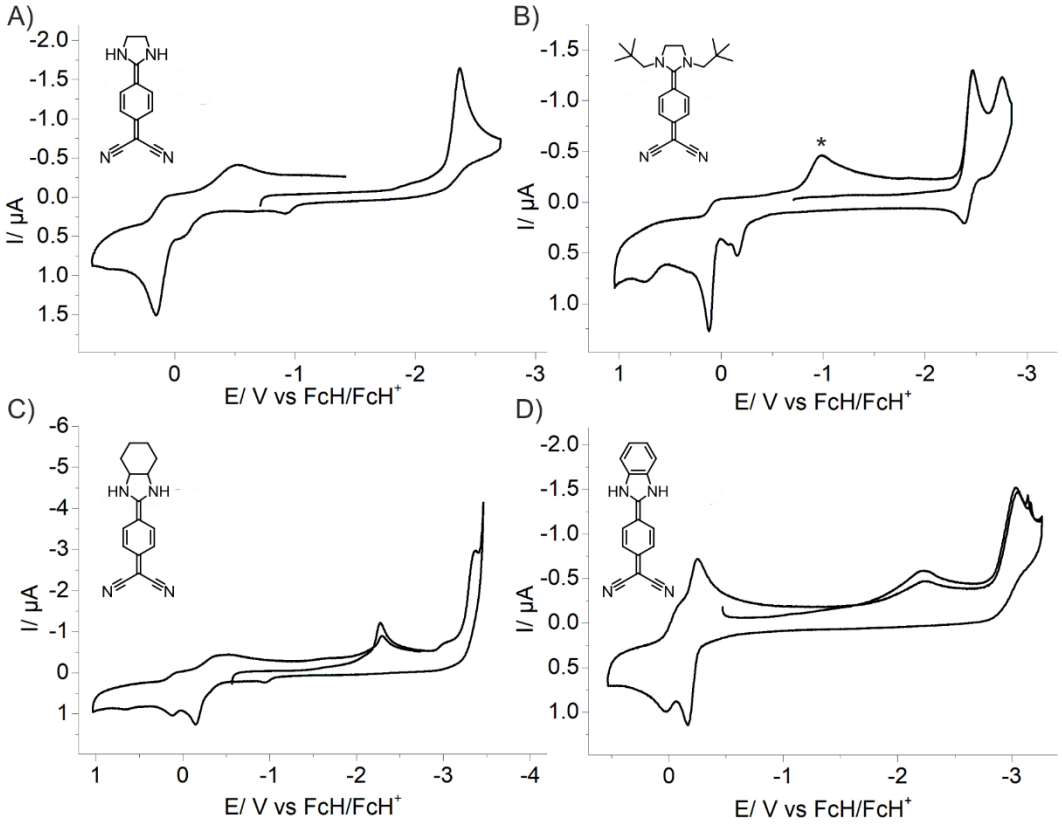


*Figure S14: Voltammograms of compound **1-4** (A-D) measured in dimethylformamide solutions with tetrabutylammonium hexafluorophosphate (0.1M). Decamethylferrocene/decamethylferrocenium ([FeCp2*] +/0) was used as the internal reference and referred to the ferrocene/ferrocenium (FcH+/0) couple. The molecular concentrations were ~ 10^{-3}. * second scan wave.*

Compounds **2-4** show two distinct reduction processes between -2 and -3.5 V (All values vs. FcH/FcH^+). Compound **1** shows only one irreversible reduction process at -2.37 V. There might be another reduction at lower potentials undetected because of the limited solvent window. Compounds **3** and **4** show two oxidations in close proximity, e.g. **4** at -0.21 and -0.07 V. Compounds **1** and **2** show only one oxidation process. Whereas the oxidations seem to be at least partly reversible for all molecules, the reduction processes differ. Compounds **3** and **4** show two irreversible processes indicating follow-up reactions (EC mechanism).[8] Compound **2** shows a reversible reduction process at -2.42 V and an irreversible one at -2.75. The single reduction of **1** seems to be irreversible and lies at -2.37 V. The differences between **1** and **2** are probably induced by the substitution at the N atoms (**2**) and the size of the N-bridging moieties (**3**, **4**).

Table S7: Redox potentials E vs FcH/FcH+ measured in DMF / 0.1 NBu4PF6 at 100 mVs^{-1} at room temperature.

	E_1 (1st Red) / V	E_2 (2nd Red) / V	E_3 (1st Ox) / V	E_4 (2nd Ox) / V
1	-2.37	---	0.16	---
2[d]	-2.42	-2.75[a]	0.12 [c]	---
3	-2.27[a]	-3.36[a]	-0.14[a,e]	0.12[a,e]
4	-2.22[a]	-3.03	-0.21	-0.07[b]
TCNQ	-0.16	-0.77	---	---

All values given were measured with GCWE at 100 mVs^{-1} given in Volts.
[a] Peak current [b] Due to overlap with 1st oxidation, potential was determined by DPV
[c] Rereduction at -0.93 V [d] Reoxidation at -0.16 V from 1st Red. and at -0.04 V and 0.74 V from 2nd Red. [e] Rereduction peaks poorly defined

Comparison to TCNQ

TCNQ shows two reversible reduction potentials around -0.06 V and -0.6 V in acetonitrile against a Ag/Ag^+ electrode.[9] Since a comparison to our measurements would lack 100% identical measurement conditions (electrode and solvent) we have measured TCNQ in DMF under the same conditions as compound 1-4. The voltammogram shows two reversible reduction processes with potentials at -0.16 V and -0.77 V.

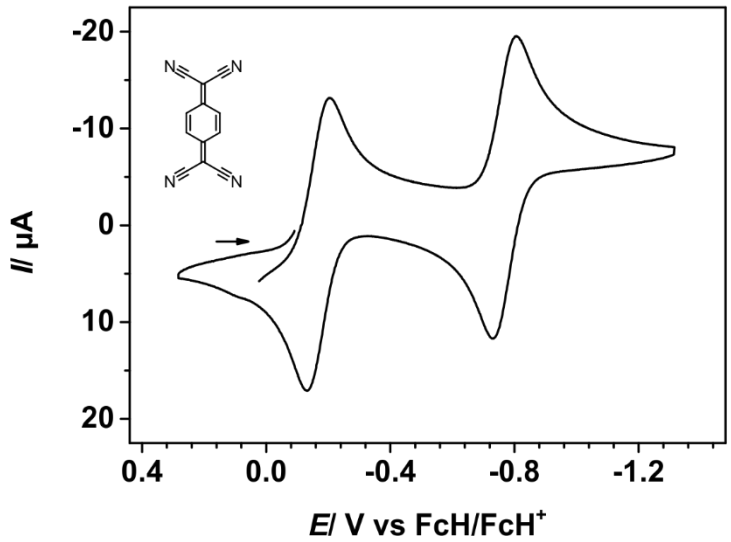


Figure S15: Voltammogram of TCNQ. measured in dimethylformamide solution with tetrabutylammonium hexafluorophosphate (0.1 M). Decamethylferrocene/decamethylferrocenium ([FeCp2] +/0) was used as the internal reference and referred to the ferrocene/ferrocenium (FcH+/0) couple. The molecular concentration was 1.37 mmol/L.*

10. Computational Details

Quantum chemical calculations have been conducted to study the cause of the drastic fluorescence enhancement in **4** as compared to the other investigated substances. Geometry optimizations and potential energy surface (PES) scans were performed using the program package ORCA (4.0.0)[10] and the stand-alone program DFT/MRCI,[11] respectively. Ground state geometry optimizations were carried out at the RI*J*COSX-CAM-B3LYP-D3(BJ)/def2-TZVP[12] level of density functional theory (DFT). Since hydrogen bonding-induced fluorescence quenching in protic solvents (MeOH, EtOH) and fluorescence enhancement by solvents with a high viscosity (DMSO) are well known phenomena for charge transfer-operated dyes and for molecules where the nonradiative excited state involves intramolecular rotations of functional moieties.[13] We opted for acetonitrile (ε = 37.5 at 20 °C)[6] as the solvent which was incorporated in the calculations with the implicit solvent model CPCM.[14] ORCA's default thresholds (e.g., 5 μHartree for energy change) were used for the calculations.

CAM-B3LYP was used as it has been shown to perform accurately for excited states and species displaying a strong charge-transfer character.[15] The structures were confirmed to be genuine minima on the PES by showing no negative (imaginary) frequencies. Excited state geometry optimizations were performed using time-dependent DFT (TD-DFT)[16] at the same level. Note that the implicit solvation cavity here is the one obtained for the ground state electronic structure. No separation of slow and fast term contributions was performed.[17] While the reduced dipole moment of the excited state (Table S8) may introduce some error for the optimization of the excited state geometry, the inclusion of state-specific solvent effects would have required the use of a different program which was not deemed necessary for our purposes.

Table S8: Oscillator strengths (length representation) for the $S_0 \rightarrow S_1$ excitation and dipole moments (given in Debye) of S_0 and S_1 for all compounds optimized in their respective electronic state at the CPCM/CAM-B3LYP-D3(BJ)/def2-TZVP level.

	Structure optimized for	**1**	**2**	**3**	**4**	**7**
$f_{osc}(S_0 \rightarrow S_1)$	S_0	1.06	0.74	1.18	1.57	1.33
	S_1	0.99	0.84	1.04	1.50	1.30
D	S_0	30.82	31.01	30.91	29.34	29.85
	S_1	25.08	24.40	24.61	23.74	22.57

Ground state *vs* excited state structures

The differences between excited state and ground state geometries are almost negligible and are only expressed in the fact that the dihedral angle D_α opens up a little (Figure S16). Test calculations showed that the inclusion of explicit solvent molecules may cause more significant changes in the structures (structures not shown). However, this is only the case if the implicit solvent model is excluded.

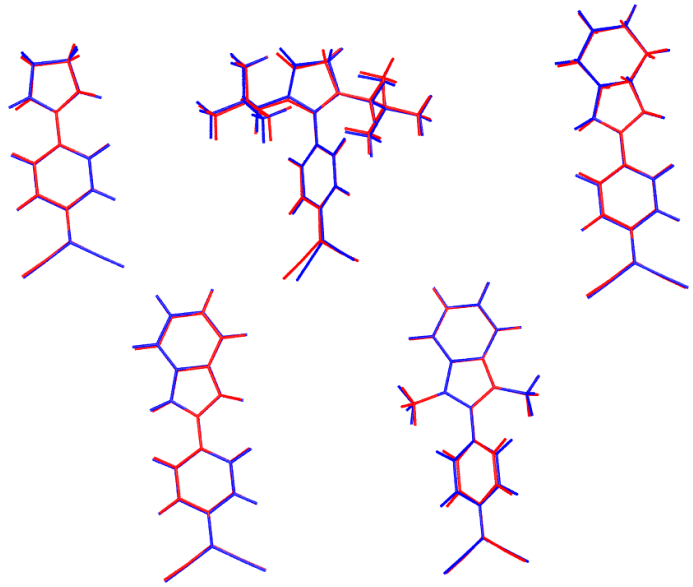


Figure S16: Comparison of the ground state (blue) and excited state (red) structures calculated at the CPCM/CAM-B3LYP-D3(BJ) level of DFT and TD-DFT, respectively.

PES scan

Obviously, the slight structural changes upon excitation do not account for the drastic differences in fluorescence QY. Therefore, we took a closer look at the PES of the intramolecular rotations around dihedral angles D_α and D_β (main text) that are likely responsible for the fluorescence deactivation. The DFT/MRCI scheme was utilized with orbitals generated at the CPCM/BHLYP/def2-TZVP[18] level as the basis to construct the configuration space. Table S9 shows the recalculated oscillator strengths (in length representation) and dipole moments of the S_0 and S_1 state evaluated at the Franck-Condon point.

Table S9: Oscillator strengths (length representation) for the $S_0 \rightarrow S_1$ coupling and dipole moments (given in Debye) of S_0 and S_1 for all compounds at the Franck-Condon point computed at the CPCM/BHLYP/def2-TZVP/MRCI level.

		1	**2**	**3**	**4**	**7**
$f_{osc}(S_0 \rightarrow S_1)$		0.95	0.74	0.96	1.26	1.20
D	S_0	31.85	35.00	33.52	32.78	33.59
	S_1	22.88	23.91	23.72	22.15	21.76

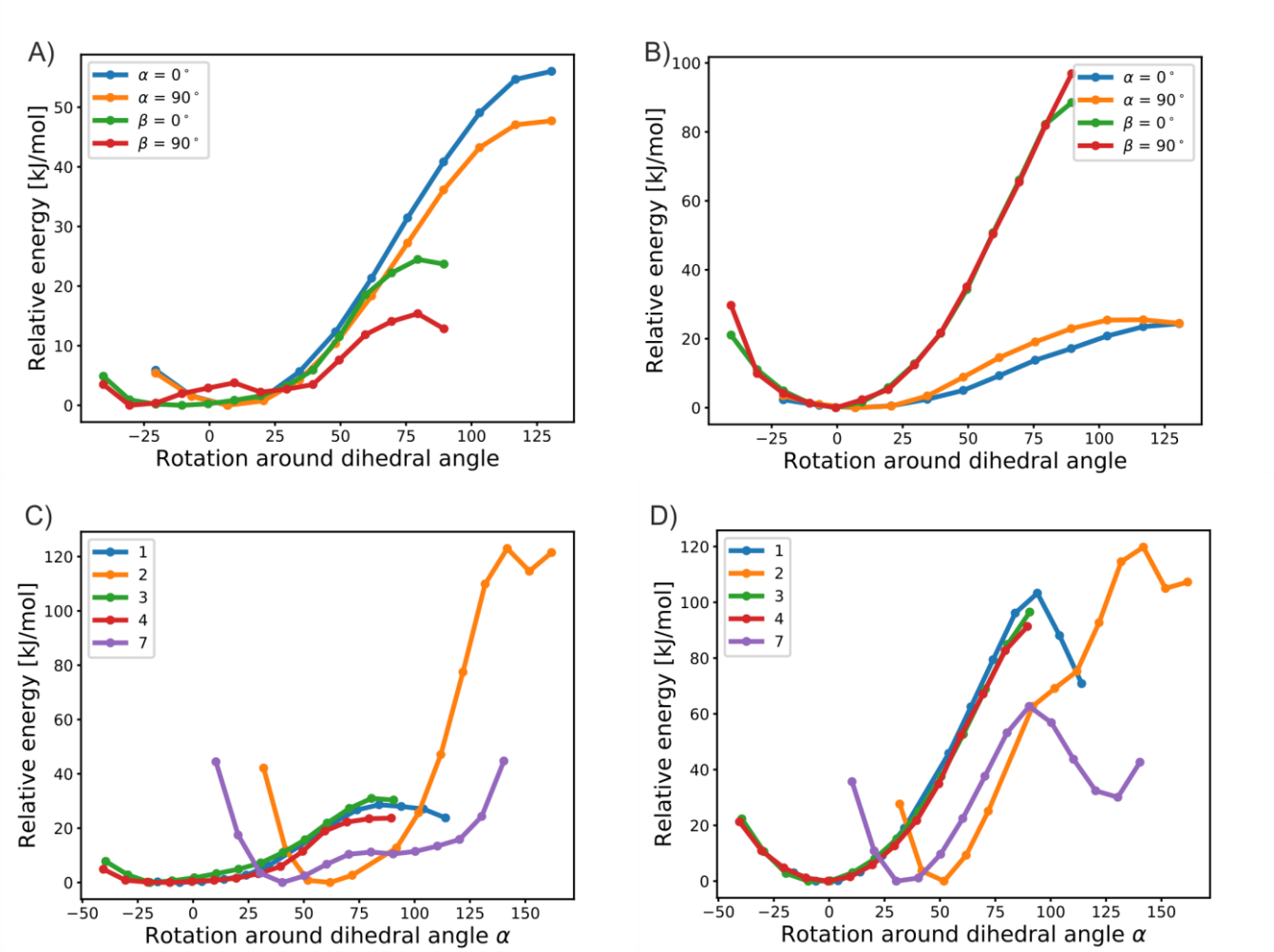


Figure S17: A) PES scan for the S0 state around dihedral angle Dα and Dβ while the respective other angle is fixed at a certain angle (see legend) for 4 computed at the CPCM/BHLYP/def2-TZVP/MRCI level. B) PES scan for the S1 state around dihedral angle Dα and Dβ while the respective other angle is fixed at a certain angle (see legend) for 4 computed at the CPCM/BHLYP/def2-TZVP/MRCI level. C) PES scan for the S0 state around dihedral angle Dα for all compounds computed at the CPCM/BHLYP/def2-TZVP/MRCI level. D) PES scan for the S1 state around dihedral angle Dα for all compounds computed at the CPCM/BHLYP/def2-TZVP/MRCI level.

Initial test calculations showed that both angles can be assumed to be decoupled (Figure S17A and B). If one angle is changed, e.g., from 0 ° to 90 °, this has a small effect on the shape of the PES of the other rotation but does almost not change the rotational barrier at all.

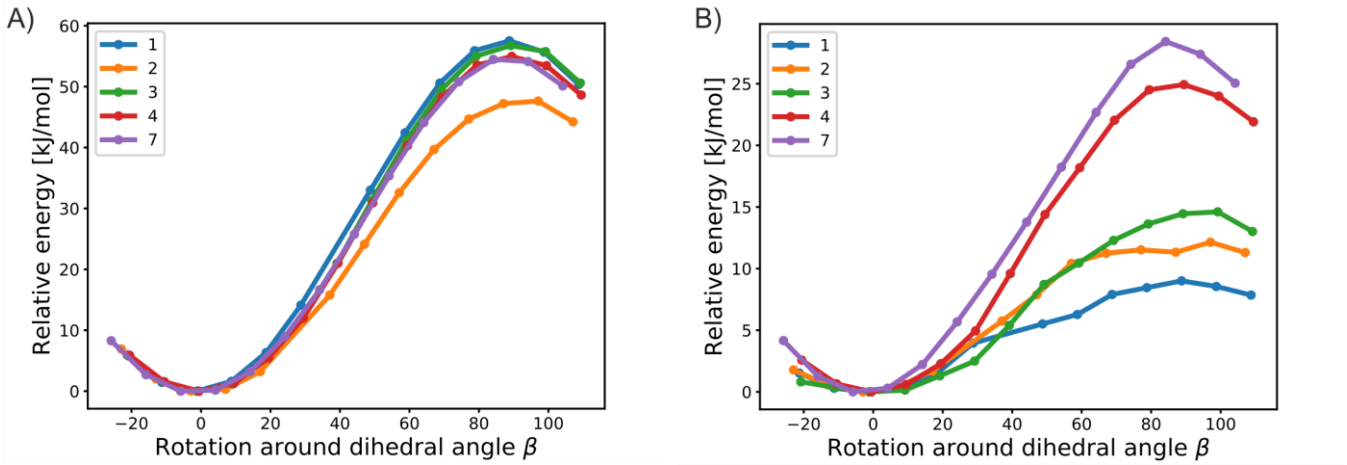


*Figure S18: A) PES scan for the S_0 state around dihedral angle D_β for all compounds computed at the CPCM/BHLYP/def2-TZVP/MRCI level. Note that for **1** and **2** values at 40 ° and 30 °, respectively, were masked as these calculations had trouble with the configuration space and converged toward an unreasonable energy. B) PES scan for the S_1 state around dihedral angle D_β for all compounds computed at the CPCM/BHLYP/def2-TZVP/MRCI level. Note that for **1** and **2** values at 40 ° and 30 °, respectively, were masked as these calculations had trouble with the configuration space and converged toward an unreasonable energy.*

Fluorescence deactivation through rotation around D_α in 1 and 7

QY loss by rotation around D_α in the S_1 can be achieved in **1** by ISC to the T_1 at an angle of 60 ° by ISC and IC events at an angle of 90 ° with the S_2, T_2 and T_3 (Figure S19 B). As it is quite unlikely for **1** to reach these displacements along D_α, this mechanism will contribute only to a small extent to the overall QY loss. However, for **7** this rotation is the most likely deactivation mechanism since it can access another local minimum on the S_0 PES of D_α at ca. 80 ° (Figure S17 C). At this point the substance may be directly photoexcited into an ISC of the S_1 with the T_1 and T_2 (Figure S19 A) drastically diminishing its chance to fluoresce.

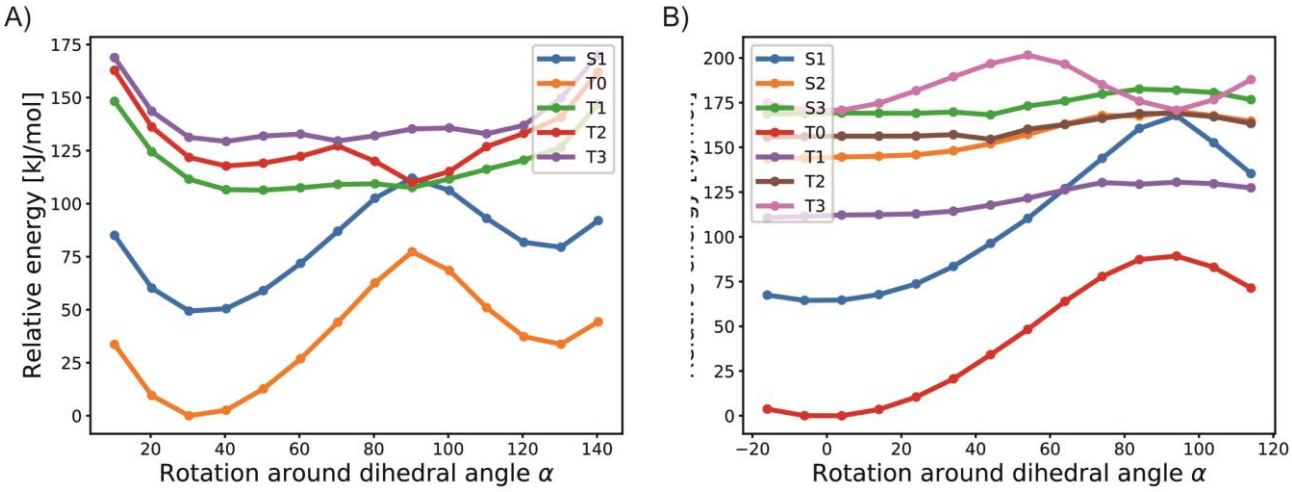


*Figure S19: A) PES scan along dihedral angle D_α for **7** computed at the CPCM/BHLYP/def2-TZVP/MRCI level to estimate fluorescence quenching through IC and ISC events. B) PES scan along dihedral angle D_α for **1** computed at the CPCM/BHLYP/def2-TZVP/MRCI level to estimate fluorescence quenching through IC and ISC events.*

Fluorescence deactivation through rotation around D_β in 1, 2, 3, and 4

To relate the probabilities of the substances to overcome a rotational barrier, we introduce the reciprocal rate constant, (Equation S5):

$$k_{\mathrm{rot}}{}^{-1} = A^{-1}\, e^{\frac{E_{\mathrm{rot}}}{\mathrm{RT}}} \qquad \text{(Equation S5)}$$

where R is the ideal gas constant, and T is the temperature (here constant at 300 K). E_{rot} is the rotational barrier height. A is the frequency factor which is approximated by the zero-point vibrational energy E_0, (Equation S6), of the harmonic potential fitted to the rotational energy profile $E(\varphi)$ with φ being the angle of rotation, (Equation S7).

$$A \sim E_0 = \tfrac{1}{2}\,\hbar\omega_0 \qquad \text{(Equation S6)}$$

$$E(\varphi) \approx \mathrm{a}\varphi^2 + \mathrm{b} \qquad \text{(Equation S7)}$$

The potential's second derivative gives its curvature, (Equation S8), which is related to the force constant of the potential, (Equation S9).

$$\frac{\partial^2 E}{\partial \varphi^2} = 2\mathrm{a} = k \qquad \text{(Equation S8)}$$

$$\omega_0 \sim \sqrt{k} \qquad \text{(Equation S9)}$$

It follows that the frequency factor is proportional to the square root of the curvature at the minimum of the rotational potential (Equation S10),

$$A \sim E_0 \sim \sqrt{a} \qquad \text{(Equation S10)}$$

The reader is referred to standard quantum chemistry and physical chemistry textbooks for a more profound derivation of the underlying mathematical principles.

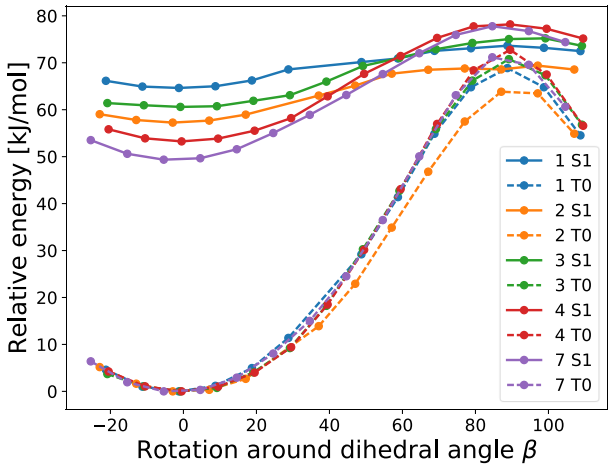


Figure S20: PES scan for the S_1 and T_0 states around dihedral angle D_β for all compounds computed at the CPCM/BHLYP/def2-TZVP/MRCI level to assess possible ISC events.

Incorporating explicit solvent effects

While the PES scans may account for the differences in QY, they do not fully explain the large QY of **4** by itself. As solvent effects have only been included in an implicit way by the continuum model CPCM, an explicit solvent cavity consisting of six ACN molecules (two near D_α and four near D_β, respectively) was set-up and the rotational barrier at D_β was recalculated for **1** and **4** with the smaller def2-SVP basis set and fixed solvent molecules to save computational resources (Figure S21A and B). The explicit solvent cavity was prepared by manually placing the solvent molecules at the desired positions and optimizing the solvent molecules with a distance constraint to the solute so that the dicyano moiety can still rotate rather freely at the CAM-B3LYP/def2-SVP level with ORCA's "loose" geometry optimization criteria. While in both cases the smooth shape of the PES naturally deteriorated, surprisingly, for **1** the total barrier hardly changed at all by less than 1 kJ/mol. **4** on the other hand shows a slight increase in barrier height by ca. 5 kJ/mol, which amounts to roughly a third of the barrier from the calculation using only an implicit solvent cavity. While these numbers are more prone to errors due to the smaller basis set, it can be deduced that the solvent shell around the molecules may play a significant role at least in the case of **4**, where it likely contributes to its large QY.

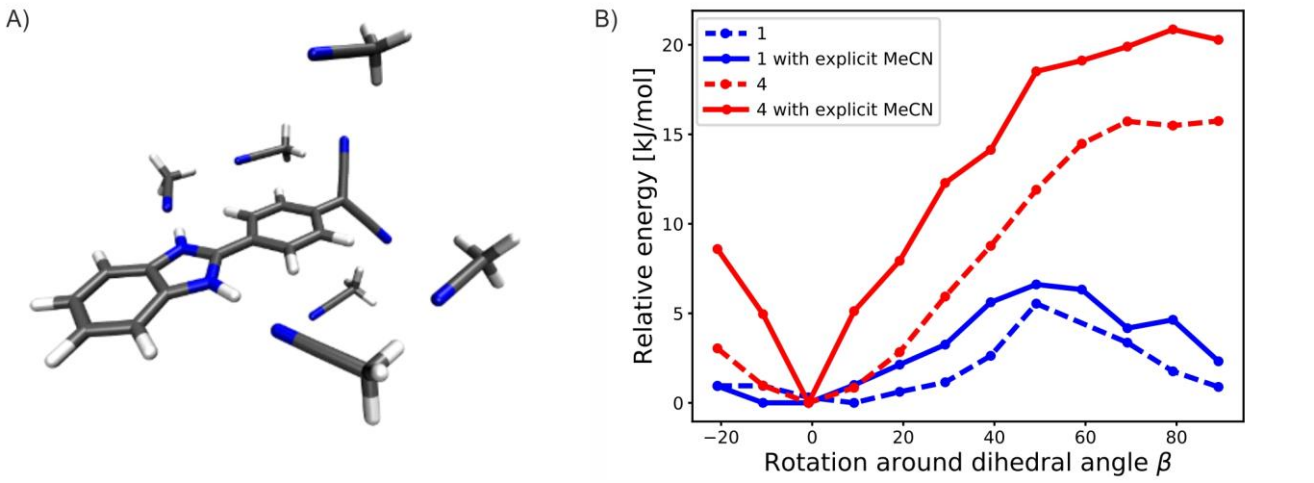


*Figure S21: A) Explicit solvent molecules placed in the vicinity of the solute to estimate solvent effects for the rotational potential at D_β. B) PES scan for the S_1 state around dihedral angle D_β for **1** and **4** with (solid) and without (dashed) six explicit ACN molecules computed at the CPCM/BHLYP/def2-SVP/MRCI level to assess the influence of solvent effects on the potential barrier.*

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